Attention and Personal Calibration for VitalDB Blood-Pressure Estimation: Bad-Blood v2

Bad-Blood Project
Research and Engineering Report

Abstract

We evaluate lightweight attention interventions on the actual high-resolution Bad-Blood CNN and early personal calibration for later contemporaneous blood- pressure estimation. Waveform models receive ten-second ECG/PPG windows and output scalar SBP/DBP. The controlled population comparison excludes previously explored personal patients and retrains the unchanged baseline. A separate six-patient chronological pilot evaluates sparse ABP-reference calibration, with adaptation settings chosen on four different patients. Results below are computed from completed, hash-verified experiments, not simulations of model accuracy. Sparse references are simulated from real invasive labels; they are not actual cuff events. The prospective four-model validation comparison retains the unchanged CNN, whose primary-test patient-macro SBP/DBP MAE, averaged across seeds 42/43/44, is 12.59/8.36 mmHg. Separately, SE-last has primary-test SBP/DBP MAE 12.386/8.002 mmHg; SE-minus-CNN changes are SBP -0.20 [-0.48, +0.07]; DBP -0.36 [-0.53, -0.18] mmHg (descriptive paired 95% patient intervals). Across the six required three-seed models, the lowest observed primary-test MAEs are SBP 12.386 mmHg for SE-last and DBP 8.002 mmHg for SE-last. These observed test comparisons do not change validation-based selection. With ten simulated personal references, three-seed mean patient-macro population and full-adaptation MAE are 12.73/7.80 and 9.24/4.66 mmHg; bias correction gives 9.02/4.84, and reference mean gives 9.01/4.04. Paired patient intervals and all individual cases qualify these descriptive comparisons.

1 Research questions and motivation

The application is a population-trained estimator that can be calibrated to one person and then use newly observed ECG/PPG to estimate their current pressure. Track A asks whether attention improves the existing architecture at matched training and data budgets. Track B asks how adaptation depth and reference budget affect later error and how performance changes over time. These hypotheses do not assume that larger models or more adaptation improve every patient. Later ECG/PPG is a legitimate input; later BP is evaluator-only. The task is not open-loop future-waveform prediction.

2 Related work

SE, ECA and CBAM motivate feature-channel and temporal gates (Hu et al., 2018; Wang et al., 2020; Woo et al., 2018). Learned feature channels are not individual sensors after convolutional mixing. Physiological hybrids offer relevant precedents but differ in calibration, scale and splitting; their reported numbers are not interchangeable benchmarks. One-reference conditioning (Schlesinger et al., 2020) and restricted-layer transfer (Leitner et al., 2022) motivate the personal comparison. Shuffled within-patient evaluation does not establish durability. Real longer-term evidence is sensor- and model-specific (Yavarimanesh et al., 2022), while strong reference-only comparators are essential (Mieloszyk et al., 2022). The two accompanying ACL literature reviews document paper-level definitions, reported metrics, provenance and limitations.

3 Data and provenance audit

Official VitalDB-specific PulseDB subsets provide the population data (Wang et al., 2023). Source suffix 1 identifies VitalDB, not a visit. All 31 case-to-subject links in complete local individual files were checked against public VitalDB metadata. The population subsets contain no timestamps or case IDs; temporal evaluation instead uses individual files with original T and CaseID. Raw ABP, derived labels and QC fields never enter the inference inputs.

Excluding all 31 historically explored patients

leaves 1,085 training patients (390,600 windows), 192 validation patients (69,120), 141 primary-test patients (56,400), and 116 stress-test patients (666). Original V1 scores are historical and are not treated as the new baseline. Previous project work inspected the test cohort, so this is a controlled rerun rather than untouched external validation. All new population weights exclude the personal patients.

Thirty complete individual files contain 30 patients, 31 cases and 15,486 windows. One truncated and one absent file are excluded; two patients have only two isolated windows. A multi-case patient is not used for the main within-case pilot. Availability-only allocation gives four development and six final patients. One initially selected patient lacked ten spaced early references and was replaced before training, with the complete audit trail retained. Final patients provide 4,284 later queries; development provides 1,855. Two query-to-query timestamp pairs overlap by 8 ms upstream; no support/query boundary is crossed. No local actual cuff/NIBP events or dated repeat visits support clinical cuff or cross-visit claims.

4 Methods and controlled design

4.1 Architecture and population fitting

The V1 source last changed at commit `1f33cdc`; the research base is `7553248`. The exact 37,666-parameter BPNet(pool=1) has a 2-to-8 stem (kernel 7, stride 2), residual blocks 8-to-16-to-32-to-48 (kernels 7 and 5), BatchNorm/SiLU, global mean pooling and a 48-to-32-to-2 head with dropout 0.15. Residual feature lengths are 313, 157 and 79. SE-last uses a 48-to-6-to-48 MLP; temporal attention uses mean/max channel pooling and a seven-tap convolution; CBAM1D applies channel then temporal gates. A 630-parameter additive pooled-context control matches SE capacity. SE-all tests placement. ECG-only/PPG-only ablations zero the absent channel. ECA is explicitly supplementary. All parameters and state keys are tested against the original implementation, rather than approximating the backbone. Figure 1 shows the shared backbone and intervention point.

Inputs use provided filtered fields with per-window/channel min–max scaling. For observed window X, the physical-unit prediction is

$$g_\theta(X) = \mu_{\mathrm{tr}} + \sigma_{\mathrm{tr}} \odot f_\theta(N(X)), \qquad (1)$$

where N scales each input channel within that window. Labels use training-only centering/scaling: $\mu_{\mathrm{tr}} = (115.418, 62.928)$ and $\sigma_{\mathrm{tr}} = (18.942, 12.083)$ mmHg, in SBP/DBP order. All fits use normalized MSE, AdamW (learning rate 0.001, weight decay 0.001), batch size 256, noise SD 0.003, gradient clipping at norm 5, maximum 40 epochs, and patience 8. A factor-0.5, patience-3 plateau scheduler and best-checkpoint selection use mean validation patient-macro SBP/DBP MAE. Seeds 42/43/44 are matched for principal comparisons; modality ablations use 42 only. Source, split, config and checkpoint hashes accompany every run. Epoch-boundary resume restores optimizer, scheduler and RNG states.

4.2 Personal calibration

Each case's first 20% of elapsed span defines the support period; query windows start after a 10-second boundary guard. Gaps are retained. From a greedy pool with starts at least 60 seconds apart, ten approximately evenly spaced references are selected; budgets 1, 2, 3, 5 and 10 use nested prefixes. Queries stay identical across budgets. Labels are invasive-ABP-derived sparse references, not cuff measurements or independent evidence merely because they are spaced.

New unchanged baseline checkpoints initialize all strategies at matched seeds. Comparisons are no adaptation, additive residual bias, shrinkage affine ($K \geq 3$), head-only, last-block-plus-head, full-model, last-reference and calibration-mean controls. The last two do not need ECG/PPG at prediction. Head, last-block and full train 1,634, 25,730 and 37,666 parameters respectively. BN running state and dropout remain fixed. Neural learning rates 0.0001/0.001 and steps 20/100, and affine penalties 1/10/100, are selected only on the four development patients averaged across seeds. No final patient's later labels control adaptation stopping, parameter selection or normalization. Bias calibration adds the support mean residual,

$$\delta_i = \frac{1}{K} \sum_{j=1}^{K} \big(y_{ij} - g_\theta(X_{ij})\big), \qquad (2)$$

to each later prediction. For each pressure separately, affine calibration fits $az_j + b$ to standardized support labels u_j by minimizing $\sum_j (u_j - az_j - b)^2 + \lambda(a-1)^2$; the intercept is unpenalized. Last-reference and reference-mean controls predict the corresponding scalar support pressure without using later waveforms. Neural adaptation minimizes the same standardized two-output MSE

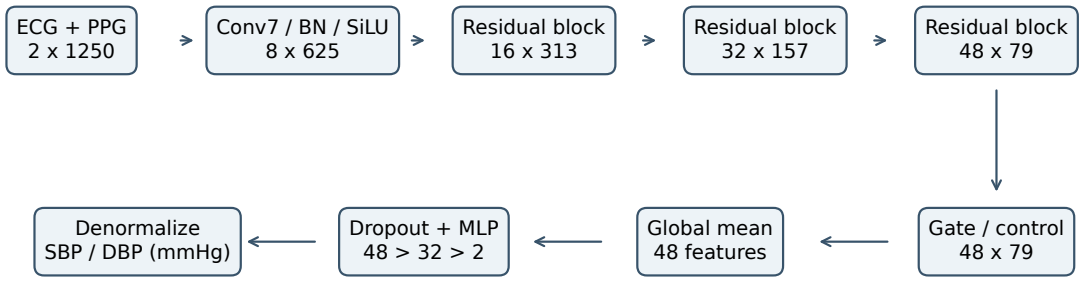


Figure 1: The unchanged high-resolution backbone and final-block intervention. Shapes omit batch size. Inputs are observed, contemporaneous ECG/PPG; outputs are scalar pressures.

on all K support windows for the development-selected number of updates; it never monitors later BP loss. The nested chronological prefixes vary both reference count and reference recency. Budget curves therefore describe this acquisition policy, not pure sample complexity with reference age held constant. Figure 2 separates calibration, later inference and evaluation.

4.3 Evaluation and uncertainty

MAE/RMSE, bias, sample error SD, median and 90th/95th error quantiles, within-5/10/15-mmHg fractions, correlation and R^2 are saved separately for SBP and DBP. Patient-macro metrics weight people equally; pooled metrics weight windows equally. Paired changes average matched seeds within patient, then resample patients 2,000 times for descriptive 95% intervals. Seed SD and patient uncertainty are separate. Reference-only predictors have identical predictions across seeds by construction; zero seed SD does not imply zero patient uncertainty or three independent replications. Counts of patients with lower error describe its sign, not a minimum practically important change. Pressure strata and all patient failures remain visible. Repeated-measurement Bland–Altman summaries decompose between/within-person error; serial correlation remains a limitation. Drift uses time from the actual last support reference, not silently the 20% boundary. Empty bins are unavailable, not zero error. The saved secondary boundary-relative analysis uses a distinct label. For one pressure and errors $e_{ij} = \hat{y}_{ij} - y_{ij}$, the primary patient-macro definitions are

$$\mathrm{MAE}_{\mathrm{macro}} = \frac{1}{P}\sum_i \frac{1}{n_i}\sum_j |e_{ij}|, \tag{3}$$

$$\mathrm{RMSE}_{\mathrm{macro}} = \frac{1}{P}\sum_i \sqrt{\frac{1}{n_i}\sum_j e_{ij}^2}. \tag{4}$$

Pooled RMSE instead takes one square root after averaging all window errors. Error SD uses denominator $N - 1$; threshold fractions include equality. Correlations and R^2 with zero required variance are undefined, not zero; macro summaries retain the number of patients for whom each metric exists.

5 Results

Figure 3 summarizes population errors. Figures 6 and 7 show the personal-reference budget and elapsed-time analyses; the paired time-bin effects are in Figure 10. Individual paired changes are in Figure 8, with chronological examples in Figure 9 and observed input signals in Figure 12. Figure 11 shows descriptive agreement. Appendix A provides complete paired, pressure-range, budget and patient-level tables. The unchanged CNN ranks first on validation among all six required three-seed architectures, including capacity and placement controls. Its primary-test patient-macro SBP/DBP MAE, averaged across seeds 42/43/44, is 12.59/8.36 mmHg. The four-model attention ranking is reported separately below; neither ranking uses test accuracy. Unless explicitly labeled seed 42, principal-model results average three separately evaluated checkpoints, not one seed-42 run or an ensemble of predictions.

At K=10, Bias changes SBP/DBP MAE by -3.71/-2.96 mmHg; 4/6 patients improve in SBP

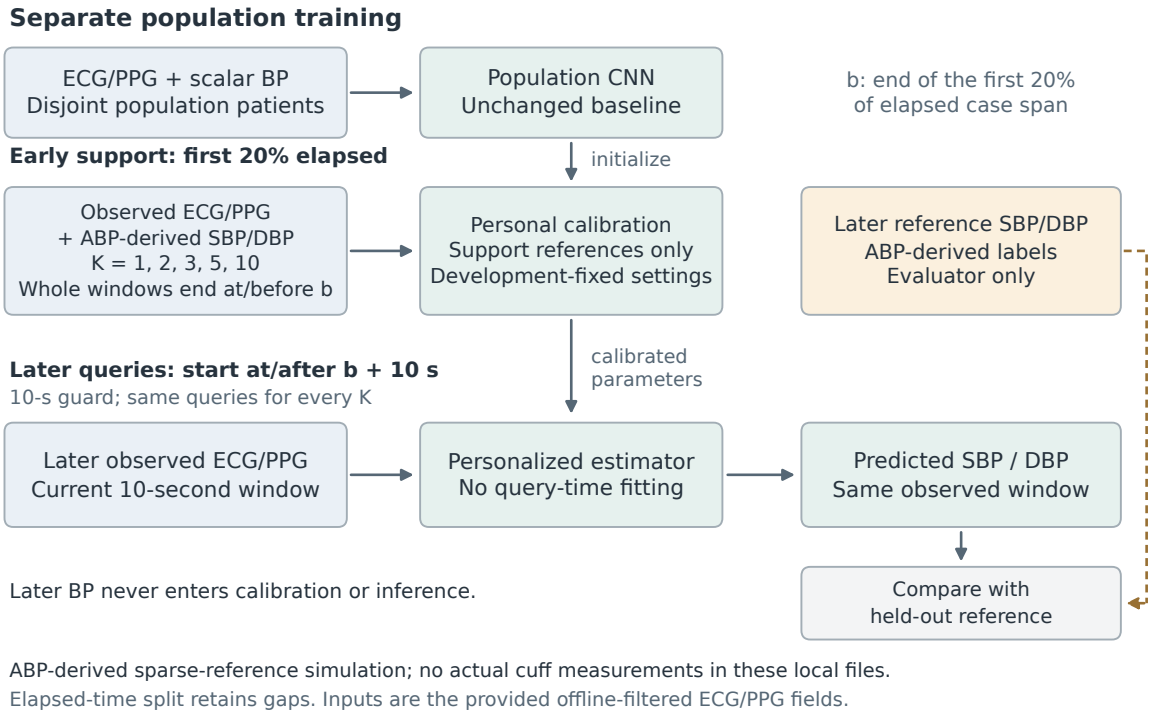


Figure 2: Personal calibration and later contemporaneous inference. Only early invasive-ABP-derived scalar references fit the personal parameters. Later observed ECG/PPG enters the estimator; later blood-pressure labels enter only the evaluator. The 20% split uses elapsed case span and preserves recording gaps.

and 4/6 in DBP.

At K=10, Head changes SBP/DBP MAE by -2.27/-2.84 mmHg; 4/6 patients improve in SBP and 5/6 in DBP.

At K=10, Last block changes SBP/DBP MAE by -3.46/-3.05 mmHg; 5/6 patients improve in SBP and 5/6 in DBP.

At K=10, Full changes SBP/DBP MAE by -3.49/-3.14 mmHg; 5/6 patients improve in SBP and 5/6 in DBP.

Validation choices. Within the four-model attention comparison (CNN, SE-last, Temporal and CBAM1D), validation selects CNN (mean SBP/DBP patient-macro MAE 10.37 mmHg). The separate required three-seed comparison, which also includes Additive and SE-all, selects CNN (10.37 mmHg). The choices coincide. Both choices use validation only; capacity, placement and modality contrasts retain their distinct roles, and optional ECA is supplementary. Table 1 separates validation SBP/DBP errors from the combined selection scores. A run is one architecture at one seed: its checkpoint is selected using that run's validation history. An architecture is ranked by the mean validation score of its three selected runs. Reported three-seed metrics average separately evaluated checkpoints, not predictions in an ensemble; a seed-42 score is one run, not the three-seed mean.

Attention effects and heterogeneity. The following are primary-test MAE changes with descriptive paired 95% patient intervals, in mmHg, relative to the same-seed CNN. Negative values reduce error. SBP and DBP remain separate rather than being replaced by the selection scalar.

SE-last: SBP -0.20 [-0.48, +0.07]; DBP -0.36 [-0.53, -0.18]. SBP/DBP macro MAE is lower in 3/3 and 3/3 matched seeds; seed-averaged patient MAE is lower for 81/141 and 96/141 patients. The two targets do not show a uniform directional advantage with both intervals on one side of zero; this is not evidence of equivalence.

Temporal: SBP -0.01 [-0.19, +0.19]; DBP -0.06 [-0.18, +0.07]. SBP/DBP macro MAE is lower in 1/3 and 3/3 matched seeds; seed-averaged patient MAE is lower for 71/141 and 76/141 patients. The two targets do not show a uniform directional advantage with both intervals on one side of zero; this is not evidence of equivalence.

CBAM1D: SBP -0.17 [-0.47, +0.13]; DBP -0.25 [-0.43, -0.08]. SBP/DBP macro MAE is lower in 3/3 and 3/3 matched seeds; seed-averaged patient MAE is lower for 72/141 and 86/141 patients. The

Model	SBP mean	DBP mean	Q_{42}	Q_{43}	Q_{44}	$\overline{Q}$
CNN	12.410	8.332	10.375	10.393	10.345	10.371
SE-last	12.515	8.264	10.291	10.457	10.421	10.390
Temporal	12.487	8.323	10.299	10.466	10.450	10.405
CBAM1D	12.521	8.294	10.379	10.383	10.461	10.408
Additive	12.442	8.340	10.400	10.366	10.407	10.391
SE-all	12.466	8.323	10.412	10.454	10.317	10.394

Table 1: Validation patient-macro MAE (mmHg), not primary-test error. Every run uses its own validation-selected checkpoint. SBP and DBP means average seeds 42/43/44; $Q_s = (\mathrm{MAE}_{s,\mathrm{SBP}} + \mathrm{MAE}_{s,\mathrm{DBP}})/2$ is one seed's score, and $\overline{Q} = (Q_{42} + Q_{43} + Q_{44})/3$ ranks architectures. The first four rows form the attention comparison; Additive and SE-all are capacity and placement controls in the separate six-model ranking. Values are rounded to three decimals for display only; selection uses unrounded values.

two targets do not show a uniform directional advantage with both intervals on one side of zero; this is not evidence of equivalence.

Intervals spanning zero do not establish equivalence, while intervals excluding zero remain unadjusted for the multiple architecture/endpoint contrasts. These intervals condition on the fitted seeds: the separate seed counts and SDs describe training variability. An average gain is not a guarantee for every patient or pressure range, and no minimum clinically useful effect was prespecified.

Architectural controls. SE-last minus Additive changes MAE by SBP -0.27 [-0.50, -0.05]; DBP -0.17 [-0.29, -0.06]. Both descriptive MAE intervals favor SE-last over Additive under the fixed fitting settings. Both contain 38,296 parameters, so parameter count alone cannot distinguish these specified operators. This is not proof that all attention mechanisms or all optimizers would behave similarly.

SE-all minus SE-last gives SBP +0.11 [-0.14, +0.37]; DBP +0.23 [+0.05, +0.42]. The two targets do not show a uniform directional advantage with both intervals on one side of zero; this is not evidence of equivalence. SE-all inserts gates at three residual outputs and has 374 more parameters than SE-last; the contrast combines insertion depth with capacity, not a pure relocation. The direct CBAM1D-versus-SE and CBAM1D-versus-Temporal MAE/RMSE contrasts are tabulated below. CBAM1D also changes channel pooling from SE's average-only descriptor to shared average/max descriptors, so it is not a factorial test isolating the added temporal gate.

Observed-sensor ablations. The zero-channel models are retrained baselines, not missing-sensor tests of the trained two-sensor model. At seed 42, PPG-only primary-test patient-macro SBP/DBP MAE is 12.296/7.771 mmHg, versus 12.603/8.235 for CNN and 12.366/7.949 for SE-last. PPG-only has lower observed point errors on both endpoints than both comparators in this run. These matched-seed point estimates are single-run evidence, not a three-seed superiority claim. ECG only minus CNN MAE: SBP +1.71 [+0.84, +2.52]; DBP +0.87 [+0.19, +1.57]; PPG only minus CNN MAE: SBP -0.31 [-0.79, +0.20]; DBP -0.46 [-0.84, -0.07]. ECG only removes PPG, whereas PPG only removes ECG. Their patient intervals condition on seed 42 and cannot establish seed-robust sensor importance or an attention-by-modality interaction. Learned feature-channel gates do not themselves identify physiological sensor importance.

Accuracy gains must be weighed against the separately measured parameters and CPU/MPS latency; parameter overhead alone is not measured speed or evidence of native-device readiness.

Recorded optimization histories. Figures 4 and 5 show the four-model attention comparison's complete saved histories, one curve per seed, without smoothing or extrapolation. Training MSE is dimensionless, using train-standardized targets and minibatch losses recorded during parameter updates with augmentation and training-mode layers. Validation instead shows mean patient-macro SBP/DBP MAE in mmHg after each epoch; circles mark the already-saved validation-selected epochs. These different objectives and evaluation conditions are not a directly comparable training-validation gap. In seed order 42/43/44, the actual/selected epochs are: CNN 9/17/16 completed, 1/9/8 selected; SE-last 9/9/9 completed, 1/1/1 selected; Temporal 9/13/17 completed, 1/5/9 selected; CBAM1D 9/10/9 completed, 1/2/1 selected. The plots describe the fixed runs; they do not es-

Contrast (left minus right)	Seeds	SBP change [95% CI]	DBP change [95% CI]
SE-last minus Additive capacity-matched operator	3	-0.27 [-0.50, -0.05]	-0.17 [-0.29, -0.06]
SE-all minus SE-last insertion depth and capacity	3	+0.11 [-0.14, +0.37]	+0.23 [+0.05, +0.42]
CBAM1D minus SE-last combined versus channel gate	3	+0.03 [-0.22, +0.28]	+0.11 [-0.01, +0.23]
CBAM1D minus Temporal combined versus temporal gate	3	-0.16 [-0.45, +0.12]	-0.19 [-0.37, -0.02]
ECG only minus CNN remove PPG, one seed	1	+1.71 [+0.84, +2.52]	+0.87 [+0.19, +1.57]
PPG only minus CNN remove ECG, one seed	1	-0.31 [-0.79, +0.20]	-0.46 [-0.84, -0.07]

Table 2: Direct architectural MAE contrasts on the primary test, in mmHg. Each estimate averages matched-seed differences within each of the 141 patients before a 2,000-resample patient bootstrap. Negative changes favor the left model. All intervals are descriptive, not multiplicity-adjusted or seed-population confidence intervals; single-seed modality contrasts condition on seed 42. These contrasts do not alter model selection.

Contrast (left minus right)	Seeds	SBP change [95% CI]	DBP change [95% CI]
SE-last minus Additive capacity-matched operator	3	-0.28 [-0.51, -0.06]	-0.23 [-0.35, -0.11]
SE-all minus SE-last insertion depth and capacity	3	+0.18 [-0.06, +0.43]	+0.34 [+0.15, +0.53]
CBAM1D minus SE-last combined versus channel gate	3	+0.02 [-0.23, +0.28]	+0.12 [+0.00, +0.24]
CBAM1D minus Temporal combined versus temporal gate	3	-0.22 [-0.51, +0.07]	-0.26 [-0.43, -0.08]
ECG only minus CNN remove PPG, one seed	1	+2.14 [+1.29, +2.98]	+1.15 [+0.49, +1.82]
PPG only minus CNN remove ECG, one seed	1	-0.34 [-0.82, +0.18]	-0.51 [-0.86, -0.14]

Table 3: Direct architectural RMSE contrasts on the primary test, in mmHg. Each estimate averages matched-seed differences within each of the 141 patients before a 2,000-resample patient bootstrap. Negative changes favor the left model. All intervals are descriptive, not multiplicity-adjusted or seed-population confidence intervals; single-seed modality contrasts condition on seed 42. These contrasts do not alter model selection.

tablish a causal overfitting mechanism or motivate retuning. All twenty required fits are complete before these diagnostics are produced; optional ECA histories are retained only in the companion JSON.

Budget-specific outcomes. All rankings below describe the fixed methods on final queries; they are not used to select a deployment model. Each MAE gives six patients equal weight and then averages three matched seeds. Differences are method minus comparator, so negative values favor the named neural method. Lowest neural denotes the smallest observed MAE among Head, Last block and Full, not a new tuned strategy; a tied neural rank uses the first of these prespecified names solely to display comparisons.

K=1. SBP: Full has the lowest observed MAE (9.86 mmHg); the lowest neural MAE is Full (9.86), with changes -4.67/-8.90/-8.90 mmHg versus Bias/Last reference/Reference mean, respectively. DBP: Full has the lowest observed MAE (5.91 mmHg); the lowest neural MAE is Full (5.91), with changes -0.56/-0.43/-0.43 mmHg versus Bias/Last reference/Reference mean, respectively.

K=2. SBP: Full has the lowest observed MAE (10.36 mmHg); the lowest neural MAE is Full (10.36), with changes -2.95/-4.84/-5.98 mmHg versus Bias/Last reference/Reference mean, respectively. DBP: Last block has the lowest observed MAE (5.03 mmHg); the lowest neural MAE is Last block (5.03), with changes -0.98/-2.55/-1.79 mmHg versus Bias/Last reference/Reference mean, respectively.

K=3. SBP: Full has the lowest observed MAE (10.57 mmHg); the lowest neural MAE is Full (10.57), with changes -3.10/-4.63/-5.04 mmHg versus Bias/Last reference/Reference mean, respectively. DBP: Last block has the lowest observed MAE (4.95 mmHg); the lowest neural MAE is Last block (4.95), with changes -1.27/-1.86/-1.34 mmHg versus Bias/Last reference/Reference mean, respectively.

K=5. SBP: Full has the lowest observed MAE (9.01 mmHg); the lowest neural MAE is Full (9.01), with changes -2.02/-8.55/-3.54 mmHg versus Bias/Last reference/Reference mean, respec-

tively. DBP: Full has the lowest observed MAE (4.80 mmHg); the lowest neural MAE is Full (4.80), with changes -0.99/-1.64/-0.59 mmHg versus Bias/Last reference/Reference mean, respectively.

K=10. SBP: Reference mean has the lowest observed MAE (9.01 mmHg); the lowest neural MAE is Full (9.24), with changes +0.22/-8.30/+0.23 mmHg versus Bias/Last reference/Reference mean, respectively. DBP: Reference mean has the lowest observed MAE (4.04 mmHg); the lowest neural MAE is Full (4.66), with changes -0.18/-4.10/+0.62 mmHg versus Bias/Last reference/Reference mean, respectively.

At K=10, Full minus Reference mean patient-macro MAE is SBP +0.226 [-1.114, +1.375]; DBP +0.620 [+0.129, +1.071] mmHg (descriptive paired 95% patient intervals). Positive changes favor Reference mean. Matched-seed differences are averaged within each of the six patients before resampling patients; these intervals are not adjusted for multiple comparisons and do not select a deployment method.

Whose prediction error improves or worsens at K=10? SBP patient counts (prediction error improves/worsens/is unchanged relative to population, after averaging matched seeds) are Bias 4/2/0; Affine 5/1/0; Head 4/2/0; Last block 5/1/0; Full 5/1/0; Last reference 3/3/0; Reference mean 4/2/0.

For SBP, the largest increase in prediction error with full adaptation is vitaldb:p005984: 7.84 versus 7.79 mmHg, change +0.04.

DBP patient counts (prediction error improves/worsens/is unchanged relative to population, after averaging matched seeds) are Bias 4/2/0; Affine 5/1/0; Head 5/1/0; Last block 5/1/0; Full 5/1/0; Last reference 3/3/0; Reference mean 5/1/0.

For DBP, the largest increase in prediction error with full adaptation is vitaldb:p005984: 5.75 versus 3.63 mmHg, change +2.13.

Within-case reference age at K=10. Full adaptation has lower SBP MAE than population in 4/4 nonempty reference-age bins and higher MAE in 0/4; the descriptive paired intervals lie wholly below/above zero in 2/0 bins, respectively.

Full adaptation has lower DBP MAE than population in 4/4 nonempty reference-age bins and higher MAE in 0/4; the descriptive paired intervals lie wholly below/above zero in 1/0 bins, respectively.

Patient membership and window coverage change across age bins; a difference between bins is not an isolated within-person decay effect. These six-patient intervals and outcome-specific ranks are descriptive, with no correction for choosing the lowest observed rank. Nested budgets jointly change reference count and recency. Neither the curves nor the ranks establish a universal durability time, a minimum cuff count, or cross-visit persistence.

6 Computational cost and engineering verification

The recorded environment is an Apple M5 with 24 GiB memory and PyTorch 2.14.0. Population fitting uses MPS; personal adaptation uses CPU. Measured desktop costs are not Raspberry Pi timings. Personal CPU execution overlapped later MPS population fits, and other local jobs were active during part of the suite. The saved timing ranges therefore include shared-host workload variation, rather than isolated-hardware benchmarks. Full source waveform hashes and patient selection hashes are verified. Tests cover V1 equality, attention shapes and counts, split separation, label/time alignment, support-only fitting, exact hand-computed metrics, frozen BN and parameters, and epoch-resume equality. No toy-test outputs enter the real inventory. The development and final personal phases require 1,068 and 684 completed units respectively. The personal inference CLI applies saved weights or correction coefficients without later labels. All eight K=10/seed-42 strategies for the first frozen target reproduced all 654 saved query predictions bitwise on CPU; source artifacts remained unchanged. This replay is an engineering consistency check, not another independent accuracy experiment.

7 Interpretation and limitations

Across the six required three-seed models, the lowest observed primary-test MAEs are SBP 12.386 mmHg for SE-last and DBP 8.002 mmHg for SE-last. These are descriptive endpoint-wise minima among prespecified fitted models, not a new model-selection rule or evidence of global superiority. Capacity and placement controls retain their distinct roles; supplementary ECA and single-seed sensor ablations do not enter this six-model summary. For SBP, SE-last minus CNN is -0.199 [-0.480, +0.068] mmHg. SE-last's observed SBP MAE is lower than

Method, outcome	0–15 min n=5, N=352	15–30 min n=5, N=379	30–60 min n=6, N=781	60+ min n=6, N=2772
Bias, SBP	-3.53 [-13.68, +6.62]	-4.01 [-12.11, +2.75]	-3.95 [-8.75, -0.38]	-3.13 [-8.28, +0.85]
Affine, SBP	-3.19 [-13.27, +7.02]	-3.85 [-11.25, +2.54]	-3.02 [-7.06, -0.29]	-2.81 [-7.15, +0.49]
Head, SBP	-2.50 [-11.85, +6.84]	-2.75 [-9.67, +3.68]	-0.91 [-5.09, +2.65]	-1.73 [-5.08, +1.47]
Last block, SBP	-4.06 [-12.92, +4.95]	-3.70 [-11.58, +3.72]	-3.23 [-7.61, +0.14]	-3.01 [-7.22, +0.05]
Full, SBP	-4.99 [-12.62, +2.85]	-4.28 [-12.15, +3.02]	-3.30 [-7.42, -0.52]	-2.91 [-6.45, -0.70]
Last reference, SBP	+1.30 [-9.02, +11.63]	-0.81 [-11.50, +8.69]	+3.80 [-6.21, +15.00]	+5.54 [-4.19, +15.22]
Reference mean, SBP	-3.40 [-13.09, +6.55]	-4.37 [-12.20, +1.04]	-3.52 [-8.23, +1.10]	-3.49 [-7.52, -0.59]
Bias, DBP	-4.36 [-10.52, +1.80]	-3.38 [-9.80, +1.61]	-1.15 [-6.28, +3.40]	-3.20 [-7.48, +0.19]
Affine, DBP	-4.61 [-10.88, +1.66]	-3.47 [-9.75, +1.73]	-1.85 [-6.54, +2.61]	-3.48 [-7.76, +0.10]
Head, DBP	-4.34 [-10.11, +1.42]	-3.26 [-9.39, +1.62]	-1.37 [-5.50, +2.08]	-2.82 [-5.81, -0.11]
Last block, DBP	-4.53 [-10.26, +1.19]	-3.19 [-9.52, +2.14]	-1.48 [-5.88, +2.35]	-3.20 [-6.83, -0.22]
Full, DBP	-4.77 [-10.22, +0.67]	-3.49 [-9.92, +2.07]	-1.76 [-5.97, +2.31]	-3.15 [-6.37, -0.35]
Last reference, DBP	-0.84 [-8.33, +6.69]	-0.86 [-7.85, +6.24]	+1.57 [-4.14, +7.73]	+1.17 [-5.51, +8.00]
Reference mean, DBP	-3.66 [-10.09, +2.98]	-3.98 [-10.57, +1.31]	-3.42 [-6.86, -0.15]	-4.08 [-8.10, -1.02]

Table 4: K=10 paired patient-macro MAE changes (method minus population; negative is better), in mmHg, with descriptive 95% patient-bootstrap intervals. Bins use query end minus actual last-reference end. Each header reports unique patients (n) and windows (N), not counts multiplied by seeds. Patients are paired within each bin and seed before averaging seeds. Empty bins are unavailable.

SBP, test	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
CNN	12.59	16.08	-1.81	15.96	10.45	25.88	31.20	25.60	48.11	66.60	0.56	0.26
SE-last	12.39	15.89	-1.76	15.78	10.19	25.65	31.13	26.03	49.22	67.60	0.55	0.28
Temporal	12.58	16.10	-1.83	15.99	10.39	25.89	31.44	25.57	48.40	66.68	0.56	0.26
CBAM1D	12.42	15.85	-0.29	15.84	10.27	25.53	31.11	25.64	48.88	67.30	0.55	0.29
Additive	12.66	16.17	-1.03	16.12	10.43	26.42	32.06	25.47	48.24	66.68	0.55	0.26
SE-all	12.49	16.01	-1.64	15.92	10.27	25.89	31.40	25.87	48.88	67.10	0.56	0.27
ECG only	14.31	18.21	-0.02	18.21	11.87	29.28	35.60	22.22	42.97	60.60	0.32	0.06
PPG only	12.30	15.69	-1.13	15.65	10.23	25.07	30.68	25.38	48.97	67.94	0.55	0.30

Table 5: Pooled-window SBP metrics on test, averaged across matched seeds (single seed for modality ablations). SD is sample error SD. Threshold columns are percentages. Complete per-patient and macro metrics are in the machine-readable results.

CNN's (change -0.199 mmHg); its paired interval includes or touches zero. For DBP, SE-last minus CNN is -0.359 [-0.534, -0.184] mmHg. SE-last's observed DBP MAE is lower than CNN's (change -0.359 mmHg); its paired interval is wholly below zero. All intervals compare a fixed model with CNN, not the lowest model with every competitor; they are descriptive, conditional on the fitted seeds, and not adjusted for multiple comparisons or selecting an observed minimum. An interval including zero leaves the direction uncertain without erasing the observed point estimate. A validation-selected model need not have the lowest test error, and a lower observed test error is not a deployment-selection decision.

The controlled attention intervention tests mechanisms, placement, modality availability and matched parameter count; equal capacity does not guarantee equal optimization. Validation ranking is a prospective selection rule, not a declaration that the selected architecture is globally best. The full prespecified test comparison supplies separate evidence about generalization; neither a favorable point estimate nor a validation ranking alone establishes clinical usefulness or deployment readiness. The per-patient table and calibration curves, not a single best number, determine whether deeper adaptation consistently helps. A neural strategy that fails to outperform reference-only controls adds complexity without demonstrated incremental predictive value on this pilot. High-pressure and low-pressure strata can reveal failures concealed by overall averages.

Six personal targets cannot establish a univer-

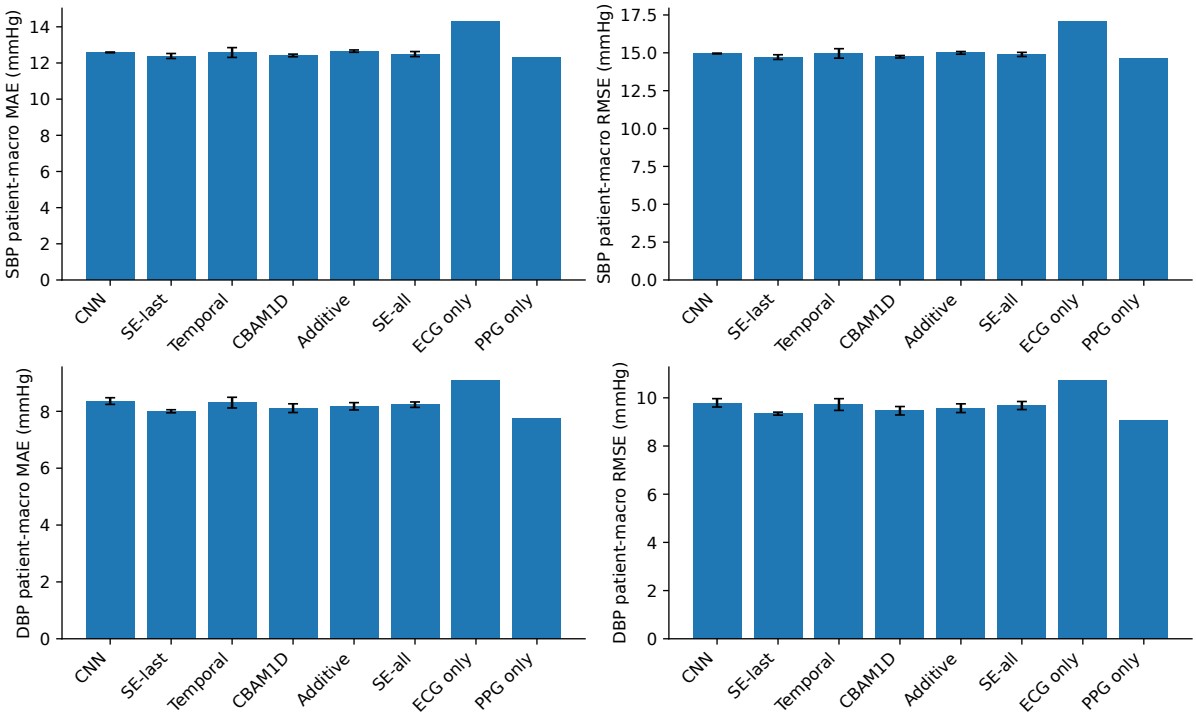


Figure 3: Primary-test patient-macro errors, averaged across seeds 42/43/44; error bars are sample seed SD. ECG-only/PPG-only results use seed 42 only and have no seed error bar. Test results do not determine selection.

DBP, test	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
CNN	8.36	10.68	-1.29	10.56	6.91	17.45	21.18	37.59	66.88	84.90	0.50	0.19
SE-last	8.00	10.19	-0.89	10.15	6.65	16.56	20.06	38.83	68.91	86.68	0.53	0.26
Temporal	8.31	10.57	-1.22	10.46	6.91	17.16	20.97	37.46	67.11	85.38	0.51	0.21
CBAM1D	8.11	10.30	-0.52	10.28	6.75	16.76	20.29	38.21	68.20	86.13	0.52	0.25
Additive	8.18	10.47	-1.00	10.42	6.72	17.05	20.80	38.63	67.95	85.59	0.51	0.22
SE-all	8.23	10.52	-1.13	10.46	6.79	17.26	20.99	38.34	67.61	85.32	0.51	0.21
ECG only	9.10	11.54	-0.01	11.54	7.53	18.95	22.98	34.39	62.76	81.27	0.31	0.05
PPG only	7.77	9.76	-1.36	9.66	6.64	15.65	19.00	38.36	69.81	88.44	0.58	0.32

Table 6: Pooled-window DBP metrics on test, averaged across matched seeds (single seed for modality ablations). SD is sample error SD. Threshold columns are percentages. Complete per-patient and macro metrics are in the machine-readable results.

sal calibration budget or durability time. The references are real invasive labels sampled to simulate sparse calibration, whereas the intended Notre Dame student measurements will use cuffs. Sensor, population and reference differences remain separate transfer questions. Provided offline filtering, upstream reference-based quality selection and an incompletely documented resampling kernel constrain online interpretation. The study evaluates later known ECG/PPG, not future waveforms. It does not test across visits or establish clinical certification. Exact negative, inconsistent and positive effects are retained in saved per-patient outputs.

8 Reproduction and artifact traceability

The review/protocol gate is commit `1b93cf1`. The repository preserves V1 and uses a dedicated research branch. Follow the README commands with Python 3.12 and the pinned requirements. Run unit tests, the population stage, then personal development and test; build this report only after the full real inventory passes hash checks. Raw recordings, detailed predictions and model weights remain outside Git under the documented artifact root. Committed JSON summaries retain run/config/checkpoint hashes and per-patient metrics.

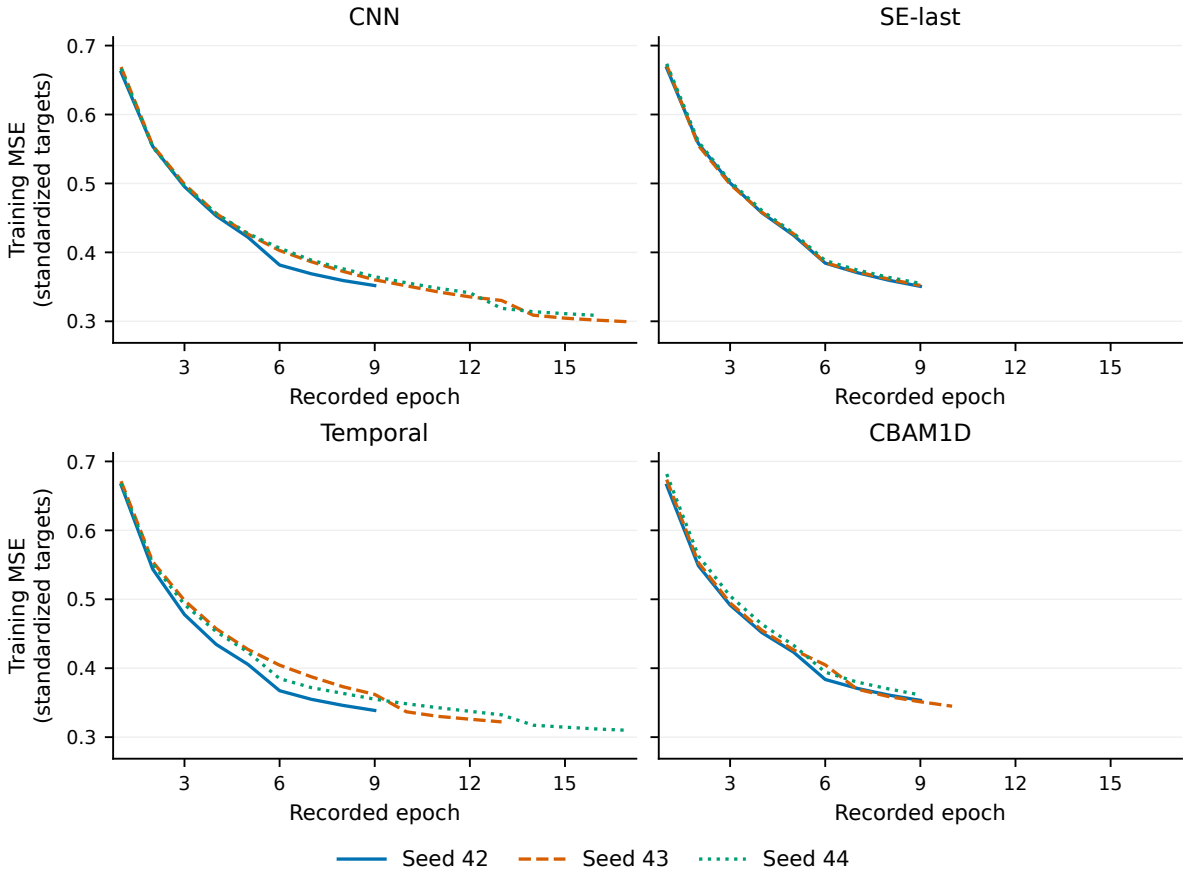


Figure 4: Complete recorded training histories for the four primary architectures. Lines distinguish seeds and stop at their actual last epochs. MSE uses standardized targets and training-time augmentation; it is not measured in mmHg.

SBP, aami	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
CNN	18.34	24.70	-11.27	21.97	13.59	40.32	53.79	19.12	37.54	54.65	0.71	0.36
SE-last	19.46	26.33	-13.22	22.76	14.43	43.08	55.30	17.82	35.79	51.75	0.69	0.27
Temporal	18.55	24.73	-11.46	21.92	14.29	39.38	53.13	18.12	36.09	51.60	0.71	0.36
CBAM1D	18.75	25.14	-11.16	22.51	14.20	40.54	53.55	18.12	35.54	52.25	0.70	0.33
Additive	18.56	25.24	-11.64	22.40	13.82	40.37	53.83	20.02	37.69	53.65	0.69	0.33
SE-all	18.57	24.80	-11.28	22.11	14.31	39.07	52.20	19.47	36.84	52.10	0.70	0.35
ECG only	25.25	32.19	-15.04	28.49	20.12	57.01	67.64	9.61	24.62	36.79	0.39	-0.09
PPG only	19.58	26.79	-12.91	23.49	14.24	44.09	57.70	20.72	37.09	52.70	0.66	0.25

Table 7: Pooled-window SBP metrics on aami, averaged across matched seeds (single seed for modality ablations). SD is sample error SD. Threshold columns are percentages. Complete per-patient and macro metrics are in the machine-readable results.

DBP, aami	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
CNN	12.15	15.46	-7.64	13.42	10.10	24.26	30.29	24.42	49.60	70.02	0.68	0.28
SE-last	12.42	16.14	-8.21	13.90	10.03	25.75	32.31	25.23	49.65	69.92	0.66	0.21
Temporal	12.32	15.72	-7.64	13.71	10.30	25.29	31.05	24.22	48.80	69.52	0.66	0.25
CBAM1D	12.06	15.61	-7.45	13.72	10.11	24.45	31.36	25.33	49.15	70.57	0.67	0.26
Additive	12.25	15.92	-8.06	13.74	10.05	25.68	32.72	25.48	49.55	71.02	0.66	0.24
SE-all	12.32	15.76	-7.53	13.85	10.52	24.91	31.35	24.02	47.95	69.72	0.65	0.25
ECG only	15.48	20.03	-10.25	17.22	12.25	33.77	37.84	23.27	43.54	58.71	0.33	-0.21
PPG only	12.85	16.33	-8.79	13.77	10.93	25.66	31.65	23.27	45.35	68.32	0.67	0.20

Table 8: Pooled-window DBP metrics on aami, averaged across matched seeds (single seed for modality ablations). SD is sample error SD. Threshold columns are percentages. Complete per-patient and macro metrics are in the machine-readable results.

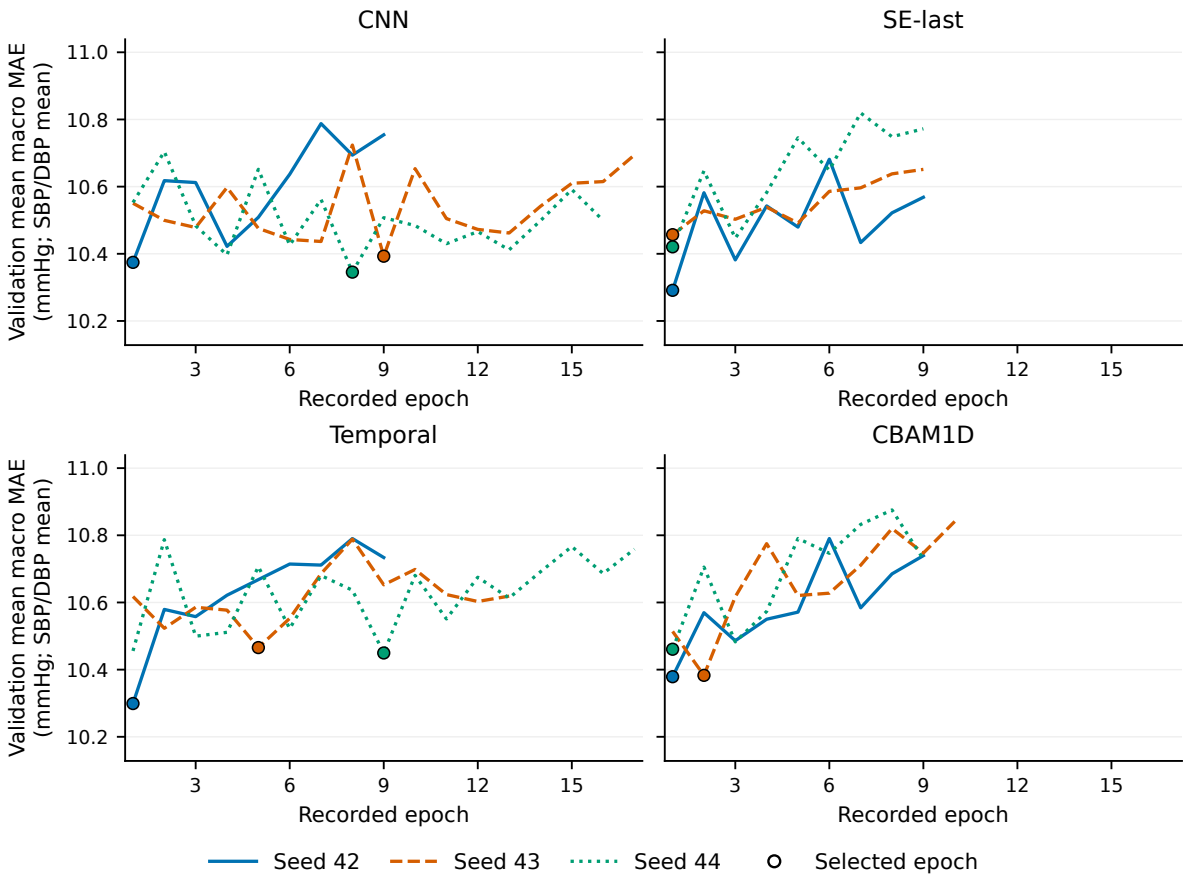


Figure 5: Complete validation histories for the same fixed fits. The ordinate averages patient-macro SBP and DBP MAE in mmHg. Circles mark saved validation-selected epochs; neither curves nor markers use test outcomes.

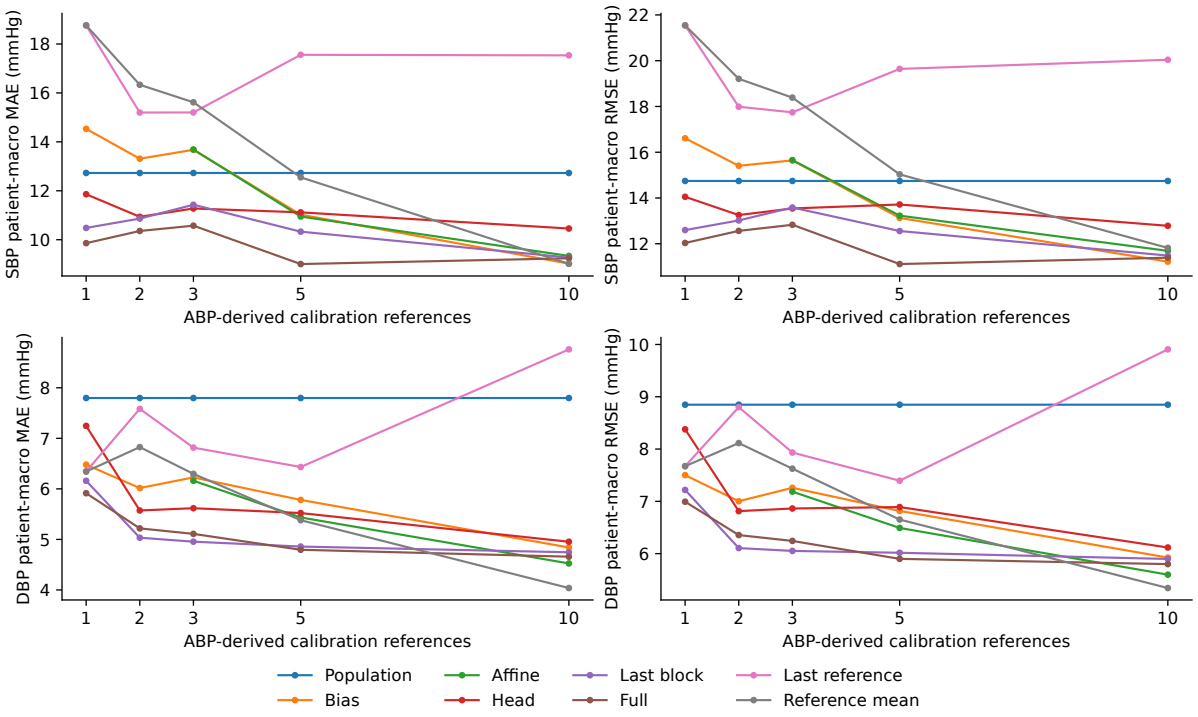


Figure 6: Calibration-budget curves on common later queries. Every curve averages six patients equally and three seeds; reference age changes with nested budget.

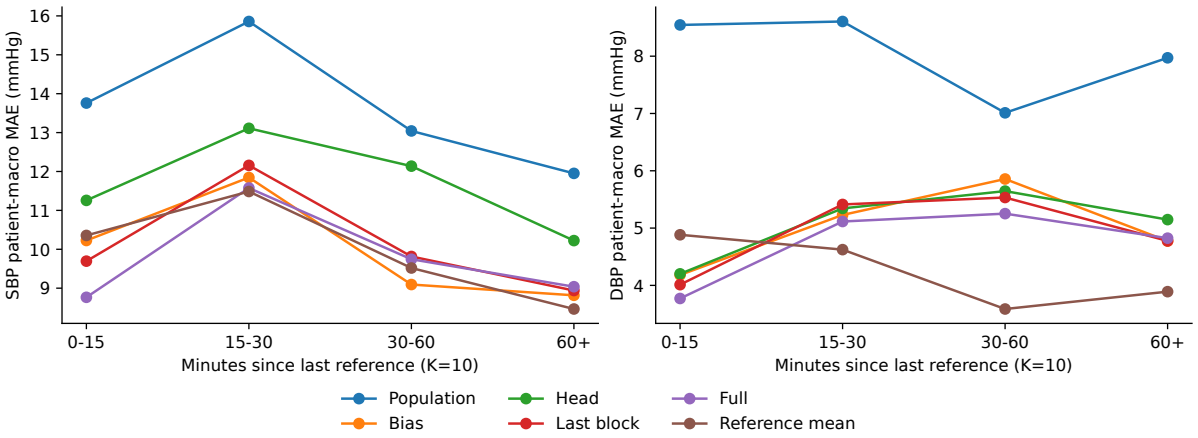


Figure 7: Within-case error versus minutes since the actual final reference at K=10. Patient counts vary by interval; complete counts are saved with the results.

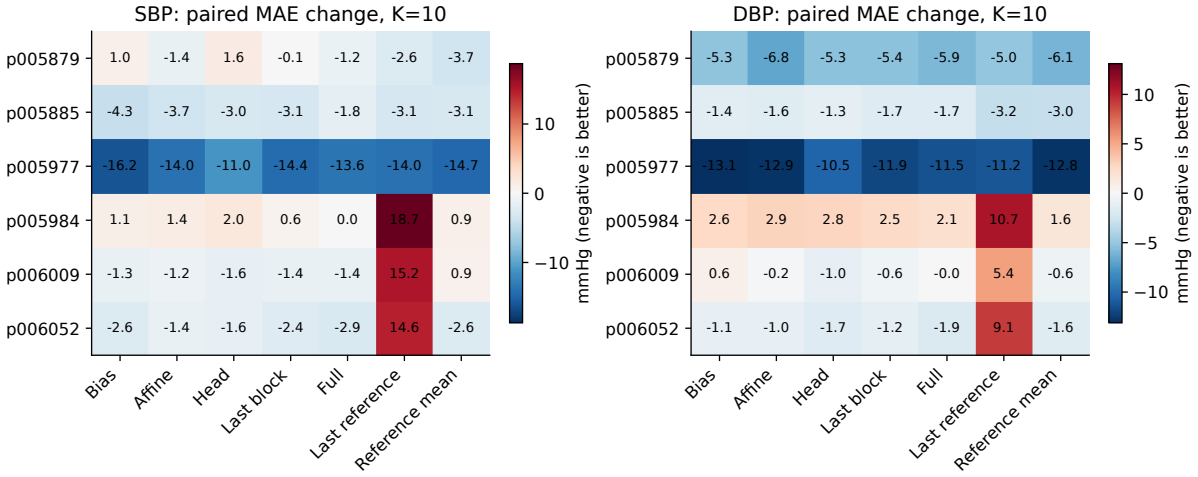


Figure 8: Every final patient's paired change at K=10. Negative values improve on the matched population model; heterogeneous failures must not be hidden.

SBP, K=10	Macro MAE	Macro RMSE	Bias	Error SD	Δ MAE (95% CI)	≤ 10
Population	12.73	14.74	3.01	15.10	Reference	43.71
Bias	9.02	11.22	-1.63	11.57	-3.71 [-9.04, 0.03]	62.43
Affine	9.34	11.69	-1.76	12.02	-3.39 [-8.02, -0.34]	61.33
Head	10.46	12.78	-1.20	13.17	-2.27 [-6.02, 0.67]	54.96
Last block	9.27	11.48	-2.39	11.71	-3.46 [-8.00, -0.45]	61.75
Full	9.24	11.39	-3.14	11.65	-3.49 [-7.65, -0.87]	60.53
Last reference	17.54	20.04	3.59	20.86	4.81 [-4.90, 14.40]	41.34
Reference mean	9.01	11.82	-0.06	12.52	-3.71 [-8.34, -0.33]	65.17

Table 9: Personal SBP results at ten simulated references. MAE/RMSE are patient-macro; Bias, Error SD and the within-10 percentage are pooled-window statistics. Entries average matched seeds. Paired changes are model minus population, averaged within patient across seeds before a 2,000-resample patient bootstrap. Intervals are descriptive for six patients.

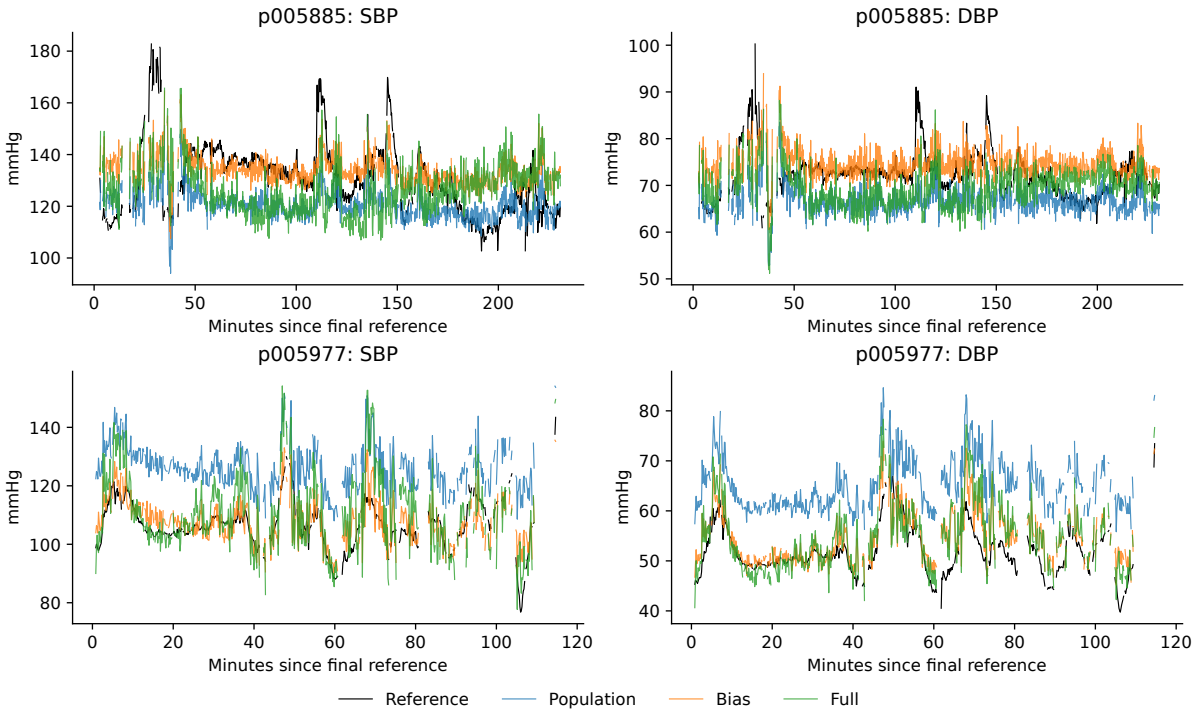


Figure 9: Representative upper-median and worst patients under baseline seed 42 query MAE, shown at K=10. Lines break at recording gaps. Bias and full-model alternatives were prespecified; no post-hoc time alignment is applied.

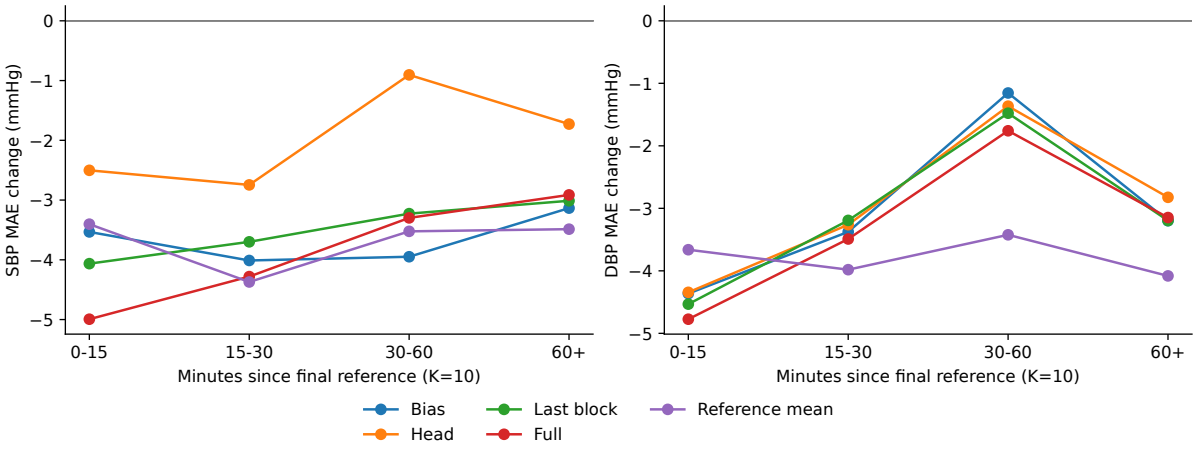


Figure 10: Paired patient-macro MAE change versus the population model in each time interval, at K=10. Negative values favor calibration. Changing patient coverage is shown in the appendix.

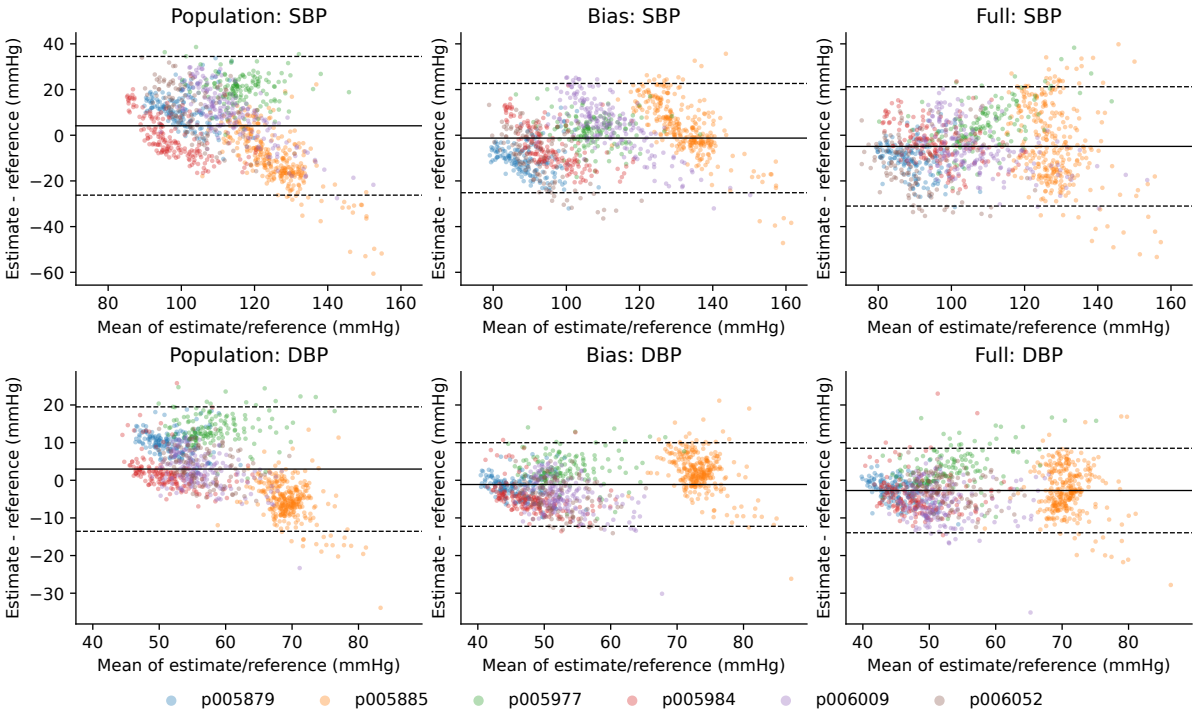


Figure 11: Descriptive repeated-measurement Bland–Altman plots for seed 42, K=10. Colors identify the six patients. Every fourth window is displayed; all windows determine the bias and random-intercept limits of agreement. Serial dependence remains, so these are not confidence bounds or clinical validation limits.

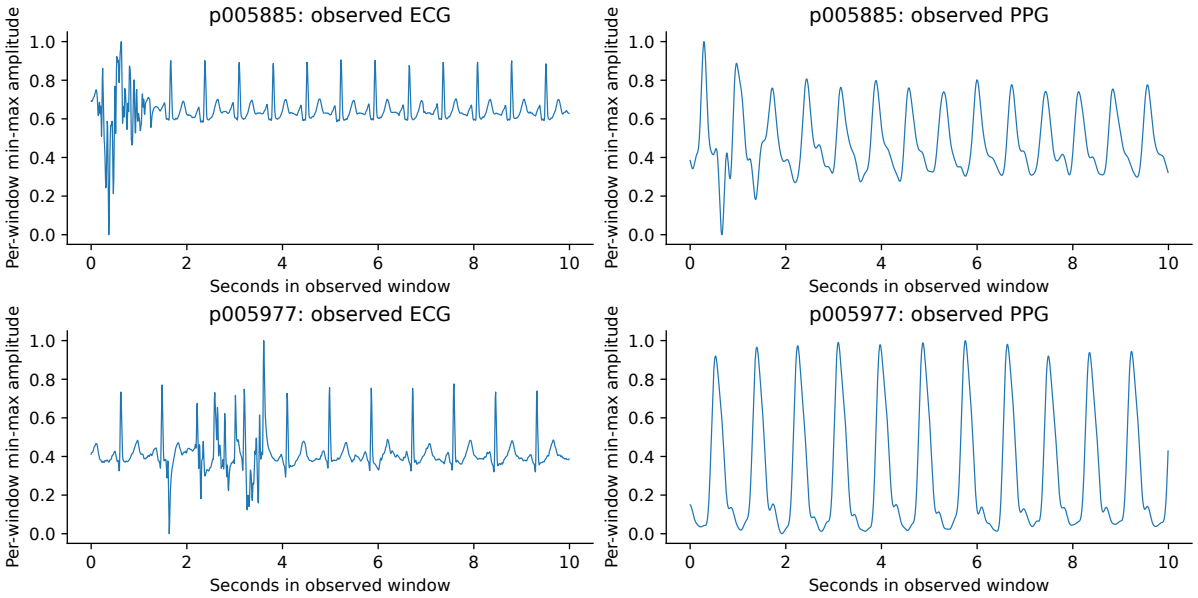


Figure 12: Actual normalized ten-second ECG/PPG input windows from the prespecified example patients. These are observed signals, not predicted future waveforms. Corresponding scalar reference and prediction values are tabulated in the appendix.

DBP, K=10	Macro MAE	Macro RMSE	Bias	Error SD	Δ MAE (95% CI)	≤ 10
Population	7.80	8.85	3.77	8.48	Reference	71.23
Bias	4.84	5.92	-1.40	6.00	-2.96 [-7.52, 0.65]	89.64
Affine	4.52	5.60	-1.02	5.77	-3.27 [-7.87, 0.52]	91.49
Head	4.95	6.12	-0.29	6.35	-2.84 [-6.53, 0.13]	89.22
Last block	4.75	5.90	-1.10	5.89	-3.05 [-7.15, 0.24]	90.30
Full	4.66	5.80	-1.07	5.77	-3.14 [-7.03, 0.10]	90.74
Last reference	8.76	9.91	1.73	9.92	0.96 [-5.45, 7.36]	66.81
Reference mean	4.04	5.34	-1.01	5.30	-3.76 [-7.85, -0.34]	92.76

Table 10: Personal DBP results at ten simulated references. MAE/RMSE are patient-macro; Bias, Error SD and the within-10 percentage are pooled-window statistics. Entries average matched seeds. Paired changes are model minus population, averaged within patient across seeds before a 2,000-resample patient bootstrap. Intervals are descriptive for six patients.

SBP, K=10	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
Population	12.66	15.42	3.01	15.10	11.45	24.22	28.15	22.46	43.71	64.01	0.57	0.29
Bias	9.14	11.69	-1.63	11.57	7.48	19.46	22.98	35.82	62.43	80.54	0.82	0.59
Affine	9.40	12.15	-1.76	12.02	7.71	19.33	23.81	33.87	61.33	80.66	0.82	0.56
Head	10.46	13.23	-1.20	13.17	9.06	20.24	24.73	28.13	54.96	76.46	0.79	0.47
Last block	9.47	11.99	-2.39	11.71	7.90	19.14	23.57	31.24	61.75	80.96	0.81	0.57
Full	9.63	12.11	-3.14	11.65	8.20	19.23	23.55	30.97	60.53	80.64	0.79	0.55
Last reference	16.55	21.16	3.59	20.86	13.58	35.31	41.77	22.46	41.34	52.99	0.38	-0.34
Reference mean	9.32	12.52	-0.06	12.52	6.96	20.69	25.65	37.79	65.17	79.18	0.75	0.53

Table 11: Complete pooled-window personal SBP metrics at K=10, averaged across three seeds. Thresholds are percentages; undefined correlations are marked unavailable. Reference-only controls may correlate when pooling different patient means, despite being constant within each patient.

DBP, K=10	MAE	RMSE	Bias	SD	MedAE	Q90	Q95	≤ 5	≤ 10	≤ 15	r	R^2
Population	7.47	9.36	3.77	8.48	6.31	15.86	18.49	41.19	71.23	88.36	0.63	0.24
Bias	4.92	6.20	-1.40	6.00	4.16	10.05	11.79	58.08	89.64	98.54	0.90	0.67
Affine	4.58	5.88	-1.02	5.77	3.79	9.54	11.48	61.85	91.49	98.67	0.89	0.70
Head	4.98	6.40	-0.29	6.35	4.12	10.19	12.27	58.35	89.22	97.88	0.88	0.65
Last block	4.75	6.09	-1.10	5.89	3.88	9.80	11.74	61.13	90.30	98.39	0.88	0.68
Full	4.71	6.03	-1.07	5.77	3.94	9.75	11.76	61.37	90.74	98.20	0.87	0.69
Last reference	7.81	10.07	1.73	9.92	6.02	17.53	19.80	43.53	66.81	83.52	0.61	0.13
Reference mean	3.96	5.40	-1.01	5.30	2.99	8.62	11.48	71.34	92.76	97.97	0.88	0.75

Table 12: Complete pooled-window personal DBP metrics at K=10, averaged across three seeds. Thresholds are percentages; undefined correlations are marked unavailable. Reference-only controls may correlate when pooling different patient means, despite being constant within each patient.

Variant	Parameters	Actual epochs	Selected epochs	Total training minutes
CNN	37,666	9,17,16	1,9,8	49.1
SE-last	38,296	9,9,9	1,1,1	26.4
Temporal	37,680	9,13,17	1,5,9	54.6
CBAM1D	38,310	9,10,9	1,2,1	31.0
Additive	38,296	9,9,11	1,1,3	41.2
SE-all	38,670	9,13,14	1,5,6	63.1
ECG only	37,666	9	1	13.0
PPG only	37,666	9	1	9.7

Table 13: Real fitting costs across included seeds. Training time includes per-epoch validation and checkpointing. Complete CPU/accelerator inference timings and memory definitions are in the result JSON.

A Complete paired, pressure-range and budget results

Model	SBP MAE	DBP MAE	SBP RMSE	DBP RMSE
CNN	12.59 ± 0.02	8.36 ± 0.12	14.96 ± 0.02	9.79 ± 0.17
SE-last	12.39 ± 0.14	8.00 ± 0.05	14.72 ± 0.15	9.34 ± 0.06
Temporal	12.58 ± 0.27	8.31 ± 0.19	14.96 ± 0.31	9.72 ± 0.24
CBAM1D	12.42 ± 0.07	8.11 ± 0.15	14.74 ± 0.08	9.46 ± 0.17
Additive	12.66 ± 0.06	8.18 ± 0.13	15.00 ± 0.09	9.57 ± 0.18
SE-all	12.49 ± 0.14	8.23 ± 0.10	14.89 ± 0.14	9.68 ± 0.17
ECG only	14.31	9.10	17.09	10.74
PPG only	12.30	7.77	14.61	9.08

Table 14: Primary-test patient-macro errors (mmHg), mean and sample SD across training seeds. The modality ablations have one seed. These SDs describe training variability, not patient uncertainty.

Nested budgets change both reference count and last-reference age: these curves are not a pure sample-complexity experiment. Primary reference-relative bins and secondary boundary-relative bins expose this distinction; no clinical minimum cuff count is inferred.

These are descriptive differences between ten sparse early ABP-derived references and all later query references, not evidence that a particular physiological mechanism or overfitting caused an error increase. SD is variability among the retained reference windows, not cuff uncertainty. Unequal sample sizes and serial dependence also affect SD and observed ranges; a wider query range alone is not proof of temporal drift.

Model minus CNN	SBP change [95% CI]	DBP change [95% CI]
SE-last	-0.20 [-0.48, 0.07]	-0.36 [-0.53, -0.18]
Temporal	-0.01 [-0.19, 0.19]	-0.06 [-0.18, 0.07]
CBAM1D	-0.17 [-0.47, 0.13]	-0.25 [-0.43, -0.08]
Additive	0.07 [-0.20, 0.32]	-0.18 [-0.34, -0.02]
SE-all	-0.09 [-0.28, 0.09]	-0.13 [-0.27, 0.00]
ECG only	1.71 [0.84, 2.52]	0.87 [0.19, 1.57]
PPG only	-0.31 [-0.79, 0.20]	-0.46 [-0.84, -0.07]

Table 15: Patient-paired MAE changes (mmHg) on the primary test. Negative values favor the intervention. Three-seed comparisons average seeds 42/43/44 within patient; ECG-only/PPG-only ablations compare seed 42 only. The bootstrap resamples 141 patients, not windows.

Model minus CNN	SBP change [95% CI]	DBP change [95% CI]
SE-last	-0.24 [-0.53, 0.02]	-0.45 [-0.62, -0.27]
Temporal	0.00 [-0.20, 0.20]	-0.07 [-0.19, 0.05]
CBAM1D	-0.21 [-0.52, 0.08]	-0.33 [-0.50, -0.15]
Additive	0.04 [-0.23, 0.29]	-0.22 [-0.39, -0.07]
SE-all	-0.06 [-0.25, 0.12]	-0.11 [-0.24, 0.02]
ECG only	2.14 [1.29, 2.98]	1.15 [0.49, 1.82]
PPG only	-0.34 [-0.82, 0.18]	-0.51 [-0.86, -0.14]

Table 16: Patient-paired RMSE changes (mmHg) on the primary test. Negative values favor the intervention. Three-seed comparisons average seeds 42/43/44 within patient; ECG-only/PPG-only ablations compare seed 42 only. The bootstrap resamples 141 patients, not windows.

SBP stratum	Windows	Patients	MAE	RMSE	Bias
CNN: below 100	12172	128	13.40	16.63	11.63
CNN: 100 to below 140	38427	141	10.62	13.30	-2.84
CNN: 140 or above	5801	113	23.86	27.53	-23.16

Table 17: Primary-test SBP pressure-range errors (mmHg), averaged across seeds. Stratum errors are pooled within each label-defined range; counts are not multiplied by seeds. Only the unchanged baseline and validation-selected architecture are shown; all models are in the JSON.

DBP stratum	Windows	Patients	MAE	RMSE	Bias
CNN: below 60	22970	135	7.70	9.90	6.00
CNN: 60 to below 90	32224	138	8.35	10.41	-5.75
CNN: 90 or above	1206	61	21.11	23.43	-21.01

Table 18: Primary-test DBP pressure-range errors (mmHg), averaged across seeds. Stratum errors are pooled within each label-defined range; counts are not multiplied by seeds. Only the unchanged baseline and validation-selected architecture are shown; all models are in the JSON.

SBP stratum, K=10	Windows	Patients	MAE	RMSE
Population: below 100	1448	5	12.80	15.18
Population: 100 to below 140	2588	6	11.25	13.61
Population: 140 or above	248	4	26.56	28.64
Bias: below 100	1448	5	7.90	9.65
Bias: 100 to below 140	2588	6	9.36	12.05
Bias: 140 or above	248	4	13.98	17.02
Head: below 100	1448	5	9.62	11.41
Head: 100 to below 140	2588	6	10.51	13.65
Head: 140 or above	248	4	14.80	17.46
Last block: below 100	1448	5	7.72	9.40
Last block: 100 to below 140	2588	6	9.76	12.35
Last block: 140 or above	248	4	16.71	19.17
Full: below 100	1448	5	6.79	8.43
Full: 100 to below 140	2588	6	10.17	12.32
Full: 140 or above	248	4	20.55	23.04
Reference mean: below 100	1448	5	6.63	8.57
Reference mean: 100 to below 140	2588	6	9.85	12.69
Reference mean: 140 or above	248	4	19.46	24.45

Table 19: Personal-query SBP pressure-range errors (mmHg), averaged across seeds. Empty reference-pressure ranges are unavailable, not perfect performance. More than one range may contain the same patient.

DBP stratum, K=10	Windows	Patients	MAE	RMSE
Population: below 60	2801	5	8.21	10.18
Population: 60 to below 90	1477	6	6.01	7.41
Population: 90 or above	6	1	21.81	22.50
Bias: below 60	2801	5	4.63	5.66
Bias: 60 to below 90	1477	6	5.45	7.04
Bias: 90 or above	6	1	13.60	14.69
Head: below 60	2801	5	4.85	6.06
Head: 60 to below 90	1477	6	5.20	6.92
Head: 90 or above	6	1	12.92	13.96
Last block: below 60	2801	5	4.62	5.73
Last block: 60 to below 90	1477	6	4.95	6.61
Last block: 90 or above	6	1	15.39	16.28
Full: below 60	2801	5	4.57	5.71
Full: 60 to below 90	1477	6	4.91	6.48
Full: 90 or above	6	1	17.64	18.47
Reference mean: below 60	2801	5	3.45	4.29
Reference mean: 60 to below 90	1477	6	4.87	6.94
Reference mean: 90 or above	6	1	19.11	19.45

Table 20: Personal-query DBP pressure-range errors (mmHg), averaged across seeds. Empty reference-pressure ranges are unavailable, not perfect performance. More than one range may contain the same patient.

Age from last reference	Windows	Patients	Pop. S	Pop. D	Bias S	Bias D	Full S	Full D
K=1, 0–15 min	0	0	–	–	–	–	–	–
K=1, 15–30 min	62	3	18.90	12.76	13.67	7.70	9.29	8.97
K=1, 30–60 min	626	6	12.42	7.05	13.05	7.30	8.99	5.34
K=1, 60+ min	3596	6	12.16	7.66	14.69	6.53	9.71	5.96
K=5, 0–15 min	33	1	10.21	4.94	22.17	6.62	15.65	6.93
K=5, 15–30 min	280	4	16.17	10.15	9.53	4.21	7.86	3.87
K=5, 30–60 min	739	6	11.75	6.84	11.93	6.77	8.22	4.43
K=5, 60+ min	3232	6	12.21	7.77	11.22	5.81	8.75	4.76
K=10, 0–15 min	352	5	13.76	8.55	10.23	4.18	8.77	3.77
K=10, 15–30 min	379	5	15.86	8.61	11.85	5.23	11.58	5.11
K=10, 30–60 min	781	6	13.04	7.01	9.09	5.86	9.74	5.25
K=10, 60+ min	2772	6	11.95	7.97	8.82	4.77	9.04	4.82

Table 21: Drift-bin coverage and patient-macro MAE (mmHg) at three budgets. S/D mean SBP/DBP. Counts are unique windows/patients, not multiplied by seeds. Differences across intervals also reflect which patients remain observed.

Patient (role)	Case	Windows	Observed min	Elapsed min	Queries	Early span min
p005548 (Dev)	c4678	584	97.33	136.33	481	27.27
p005647 (Dev)	c4746	703	117.17	240.33	486	48.07
p005968 (Dev)	c3618	498	83.00	109.50	401	21.90
p006090 (Dev)	c2134	624	104.00	161.00	487	32.20
p005879 (Test)	c5552	806	134.33	145.00	654	29.00
p005885 (Test)	c0637	1449	241.50	288.00	1220	57.60
p005977 (Test)	c3725	721	120.17	144.17	569	28.83
p005984 (Test)	c4777	797	132.83	179.83	666	35.97
p006009 (Test)	c2935	820	136.67	265.50	761	53.10
p006052 (Test)	c4462	510	85.00	100.33	414	20.07

Table 22: Chronological personal cohort. Observed duration counts retained ten-second windows; elapsed duration includes gaps. Every patient has ten available spaced reference windows; K selects a nested prefix. Queries are identical across all methods and budgets, and all ten patients were excluded from population fitting.

Patient	Source row	Reference S/D	Population	Bias	Full
p005885	229	133.91/71.62	130.33/70.96	143.74/78.67	166.14/76.28
p005977	152	98.80/45.32	122.43/57.43	103.93/46.75	90.01/40.64

Table 23: Scalar SBP/DBP values (mmHg) paired with the observed-waveform figure. These are the first later query windows of the upper-median and worst patients under seed-42 population MAE, not windows selected for a favorable prediction.

SBP MAE	K=1	K=2	K=3	K=5	K=10
Population	12.73 ± 0.74	12.73 ± 0.74	12.73 ± 0.74	12.73 ± 0.74	12.73 ± 0.74
Bias	14.53 ± 0.67	13.31 ± 0.81	13.68 ± 1.16	11.02 ± 0.82	9.02 ± 0.51
Affine	–	–	13.69 ± 1.14	10.95 ± 0.66	9.34 ± 0.39
Head	11.86 ± 0.29	10.94 ± 1.34	11.28 ± 0.99	11.12 ± 1.26	10.46 ± 0.95
Last block	10.48 ± 1.62	10.86 ± 2.53	11.43 ± 2.30	10.33 ± 1.65	9.27 ± 0.79
Full	9.86 ± 1.29	10.36 ± 2.62	10.57 ± 2.29	9.01 ± 0.95	9.24 ± 1.50
Last reference	18.76 ± 0.00	15.20 ± 0.00	15.20 ± 0.00	17.56 ± 0.00	17.54 ± 0.00
Reference mean	18.76 ± 0.00	16.33 ± 0.00	15.62 ± 0.00	12.55 ± 0.00	9.01 ± 0.00

Table 24: All-budget patient-macro SBP MAE in mmHg: mean $\pm$ sample SD across three matched population seeds, after equal weighting of the six final patients within each seed. Seed SD is not a patient confidence interval. Affine calibration is prespecified as unavailable for K below three.

SBP RMSE	K=1	K=2	K=3	K=5	K=10
Population	14.74 ± 0.67	14.74 ± 0.67	14.74 ± 0.67	14.74 ± 0.67	14.74 ± 0.67
Bias	16.61 ± 0.71	15.41 ± 0.77	15.64 ± 1.14	13.13 ± 0.73	11.22 ± 0.46
Affine	–	–	15.66 ± 1.13	13.23 ± 0.44	11.69 ± 0.36
Head	14.05 ± 0.27	13.25 ± 1.50	13.54 ± 1.22	13.72 ± 1.85	12.78 ± 1.16
Last block	12.60 ± 1.72	13.02 ± 2.52	13.58 ± 2.36	12.56 ± 1.93	11.48 ± 1.09
Full	12.03 ± 1.48	12.56 ± 2.74	12.83 ± 2.48	11.11 ± 1.09	11.39 ± 1.65
Last reference	21.54 ± 0.00	17.99 ± 0.00	17.74 ± 0.00	19.64 ± 0.00	20.04 ± 0.00
Reference mean	21.54 ± 0.00	19.21 ± 0.00	18.39 ± 0.00	15.03 ± 0.00	11.82 ± 0.00

Table 25: All-budget patient-macro SBP RMSE in mmHg: mean $\pm$ sample SD across three matched population seeds, after equal weighting of the six final patients within each seed. Seed SD is not a patient confidence interval. Affine calibration is prespecified as unavailable for K below three.

DBP MAE	K=1	K=2	K=3	K=5	K=10
Population	7.80 ± 1.11	7.80 ± 1.11	7.80 ± 1.11	7.80 ± 1.11	7.80 ± 1.11
Bias	6.48 ± 0.60	6.02 ± 0.73	6.23 ± 0.66	5.78 ± 0.82	4.84 ± 0.47
Affine	–	–	6.16 ± 0.58	5.43 ± 0.39	4.52 ± 0.11
Head	7.25 ± 0.31	5.57 ± 0.65	5.62 ± 0.58	5.52 ± 0.51	4.95 ± 0.18
Last block	6.16 ± 1.04	5.03 ± 0.47	4.95 ± 0.46	4.86 ± 0.70	4.75 ± 0.46
Full	5.91 ± 1.25	5.22 ± 0.43	5.11 ± 0.11	4.80 ± 0.35	4.66 ± 0.41
Last reference	6.34 ± 0.00	7.58 ± 0.00	6.82 ± 0.00	6.43 ± 0.00	8.76 ± 0.00
Reference mean	6.34 ± 0.00	6.83 ± 0.00	6.30 ± 0.00	5.38 ± 0.00	4.04 ± 0.00

Table 26: All-budget patient-macro DBP MAE in mmHg: mean $\pm$ sample SD across three matched population seeds, after equal weighting of the six final patients within each seed. Seed SD is not a patient confidence interval. Affine calibration is prespecified as unavailable for K below three.

DBP RMSE	K=1	K=2	K=3	K=5	K=10
Population	8.85 ± 1.07	8.85 ± 1.07	8.85 ± 1.07	8.85 ± 1.07	8.85 ± 1.07
Bias	7.50 ± 0.68	7.00 ± 0.74	7.26 ± 0.70	6.82 ± 0.84	5.92 ± 0.51
Affine	–	–	7.19 ± 0.62	6.49 ± 0.48	5.60 ± 0.10
Head	8.38 ± 0.44	6.81 ± 0.66	6.86 ± 0.59	6.89 ± 0.57	6.12 ± 0.19
Last block	7.22 ± 1.16	6.11 ± 0.54	6.05 ± 0.52	6.02 ± 0.74	5.90 ± 0.54
Full	6.99 ± 1.33	6.35 ± 0.49	6.24 ± 0.19	5.90 ± 0.43	5.80 ± 0.45
Last reference	7.67 ± 0.00	8.80 ± 0.00	7.93 ± 0.00	7.39 ± 0.00	9.91 ± 0.00
Reference mean	7.67 ± 0.00	8.11 ± 0.00	7.62 ± 0.00	6.65 ± 0.00	5.34 ± 0.00

Table 27: All-budget patient-macro DBP RMSE in mmHg: mean $\pm$ sample SD across three matched population seeds, after equal weighting of the six final patients within each seed. Seed SD is not a patient confidence interval. Affine calibration is prespecified as unavailable for K below three.

Method	Neural weights	Scalar coefficients	Enabled neural modules
Population	0	0	None
Bias	0	2	None
Affine	0	4	None
Head	1,634	0	Two head linear layers
Last block	25,730	0	Final residual block and head
Full	37,666	0	Entire CNN; BN state fixed
Last reference	0	2	None
Reference mean	0	2	None

Table 28: Personal adaptation capacity. Scalar coefficients are offset/reference values (two), or per-target slope/intercept (four); they are not extra neural weights. BN affine parameters train only inside enabled blocks. BN running statistics and dropout remain fixed; exact named masks and actual tensor changes are saved in personal_details.json.

Method, K=10	Update/fit seconds	Total adaptation seconds	Inference ms/query
Population	0.0001	0.0001	1.3343
Bias	0.0001	0.0001	1.3343
Affine	0.0001	0.0001	1.3343
Head	0.5473	0.5542	1.3217
Last block	0.7539	0.7612	1.3474
Full	1.3831	1.3909	1.2352
Last reference	0.0001	0.0001	0.0001
Reference mean	0.0001	0.0001	0.0001

Table 29: Actual local-workstation costs at K=10: median across 18 patient/seed runs. Neural update time excludes setup; total adaptation includes cloning, setup and audit. Scalar rows time calibration arithmetic, not reference acquisition. Population performs only an identity transform, not a fit. Inference is batched elapsed time divided by query count, including data transfers, not single-window latency or Raspberry Pi timing. Population inference is measured once per patient/seed and reused by bias/affine, so these rows must not be summed as independent total-run costs. Reference-only controls exclude neural inference. Complete unit measurements and ranges are saved in personal_details.json.

Method minus population	K=1	K=2	K=3	K=5	K=10
Bias	1.80 [-4.19, 8.46]	0.58 [-3.63, 4.93]	0.95 [-4.65, 7.08]	-1.70 [-7.50, 3.45]	-3.71 [-9.04, 0.03]
Affine	–	–	0.96 [-4.67, 7.15]	-1.78 [-7.59, 3.24]	-3.39 [-8.02, -0.34]
Head	-0.87 [-2.61, 1.69]	-1.79 [-6.27, 0.99]	-1.45 [-6.49, 2.57]	-1.61 [-5.80, 1.76]	-2.27 [-6.02, 0.67]
Last block	-2.25 [-6.34, 0.91]	-1.86 [-5.86, 0.78]	-1.30 [-6.19, 2.18]	-2.40 [-7.33, 0.96]	-3.46 [-8.00, -0.45]
Full	-2.87 [-6.52, -0.06]	-2.37 [-6.05, 0.34]	-2.15 [-5.93, 0.46]	-3.72 [-8.24, -0.67]	-3.49 [-7.65, -0.87]
Last reference	6.03 [0.22, 13.88]	2.47 [-0.79, 6.70]	2.47 [-4.29, 11.11]	4.83 [-3.87, 14.75]	4.81 [-4.90, 14.40]
Reference mean	6.03 [0.22, 13.88]	3.60 [-0.14, 9.66]	2.89 [-1.14, 9.63]	-0.18 [-4.77, 4.78]	-3.71 [-8.34, -0.33]

Table 30: All-budget paired SBP MAE changes in mmHg: estimate above descriptive 95% patient-bootstrap interval. For each patient, differences are averaged over the three matched seeds before resampling the six patients 2,000 times. Negative changes favor the method. These are not seed-SD intervals or simultaneous multiplicity-adjusted intervals. Affine calibration is unavailable below three references.

Method minus population	K=1	K=2	K=3	K=5	K=10
Bias	-1.32 [-5.07, 2.11]	-1.78 [-7.32, 3.51]	-1.57 [-7.02, 2.99]	-2.02 [-7.18, 2.14]	-2.96 [-7.52, 0.65]
Affine	–	–	-1.64 [-7.09, 2.89]	-2.36 [-7.42, 1.77]	-3.27 [-7.87, 0.52]
Head	-0.55 [-1.64, 0.79]	-2.23 [-5.41, 0.76]	-2.18 [-5.50, 0.58]	-2.28 [-5.55, 0.40]	-2.84 [-6.53, 0.13]
Last block	-1.64 [-3.42, -0.11]	-2.76 [-6.76, 0.45]	-2.84 [-6.85, 0.29]	-2.94 [-7.00, 0.29]	-3.05 [-7.15, 0.24]
Full	-1.88 [-3.78, -0.48]	-2.58 [-6.20, 0.20]	-2.69 [-6.53, 0.26]	-3.00 [-7.13, 0.22]	-3.14 [-7.03, 0.10]
Last reference	-1.45 [-6.12, 3.19]	-0.21 [-3.11, 2.56]	-0.98 [-4.66, 2.96]	-1.37 [-6.22, 2.25]	0.96 [-5.45, 7.36]
Reference mean	-1.45 [-6.12, 3.19]	-0.97 [-4.57, 2.54]	-1.50 [-5.10, 2.22]	-2.42 [-5.61, 0.57]	-3.76 [-7.85, -0.34]

Table 31: All-budget paired DBP MAE changes in mmHg: estimate above descriptive 95% patient-bootstrap interval. For each patient, differences are averaged over the three matched seeds before resampling the six patients 2,000 times. Negative changes favor the method. These are not seed-SD intervals or simultaneous multiplicity-adjusted intervals. Affine calibration is unavailable below three references.

Method minus population	K=1	K=2	K=3	K=5	K=10
Bias	1.87 [-4.10, 8.47]	0.66 [-3.42, 4.89]	0.90 [-4.46, 6.83]	-1.61 [-7.22, 3.44]	-3.53 [-8.66, 0.04]
Affine	–	–	0.91 [-4.46, 6.90]	-1.52 [-7.06, 3.36]	-3.05 [-7.33, -0.23]
Head	-0.69 [-2.28, 1.87]	-1.49 [-5.52, 1.08]	-1.20 [-5.82, 2.56]	-1.03 [-4.82, 2.06]	-1.96 [-5.27, 0.63]
Last block	-2.15 [-5.98, 0.92]	-1.72 [-5.50, 0.87]	-1.16 [-5.86, 2.28]	-2.19 [-6.90, 1.06]	-3.26 [-7.50, -0.43]
Full	-2.71 [-6.09, -0.04]	-2.18 [-5.72, 0.48]	-1.91 [-5.54, 0.58]	-3.63 [-7.83, -0.69]	-3.35 [-7.16, -0.89]
Last reference	6.80 [0.94, 14.46]	3.24 [-0.16, 7.43]	3.00 [-3.62, 11.24]	4.90 [-3.60, 14.29]	5.30 [-4.61, 14.71]
Reference mean	6.80 [0.94, 14.46]	4.46 [0.62, 10.32]	3.64 [-0.41, 10.09]	0.29 [-4.28, 5.33]	-2.93 [-7.70, 0.83]

Table 32: All-budget paired SBP RMSE changes in mmHg: estimate above descriptive 95% patient-bootstrap interval. For each patient, differences are averaged over the three matched seeds before resampling the six patients 2,000 times. Negative changes favor the method. These are not seed-SD intervals or simultaneous multiplicity-adjusted intervals. Affine calibration is unavailable below three references.

Method minus population	K=1	K=2	K=3	K=5	K=10
Bias	-1.35 [-4.92, 1.82]	-1.85 [-7.05, 3.21]	-1.59 [-6.71, 2.81]	-2.03 [-6.89, 1.98]	-2.92 [-7.21, 0.45]
Affine	–	–	-1.66 [-6.78, 2.72]	-2.36 [-7.09, 1.57]	-3.25 [-7.52, 0.28]
Head	-0.47 [-1.44, 0.68]	-2.04 [-4.85, 0.61]	-1.99 [-4.84, 0.43]	-1.96 [-4.84, 0.41]	-2.73 [-6.02, -0.02]
Last block	-1.63 [-3.24, -0.30]	-2.74 [-6.53, 0.27]	-2.80 [-6.61, 0.13]	-2.83 [-6.55, 0.20]	-2.95 [-6.73, 0.07]
Full	-1.86 [-3.54, -0.60]	-2.49 [-5.89, 0.13]	-2.61 [-6.18, 0.19]	-2.95 [-6.83, 0.08]	-3.05 [-6.55, -0.09]
Last reference	-1.18 [-5.71, 3.32]	-0.05 [-3.03, 2.75]	-0.92 [-4.37, 2.91]	-1.45 [-5.82, 1.93]	1.06 [-4.92, 7.20]
Reference mean	-1.18 [-5.71, 3.32]	-0.73 [-4.10, 2.65]	-1.22 [-4.71, 2.44]	-2.20 [-5.21, 0.73]	-3.51 [-7.38, -0.25]

Table 33: All-budget paired DBP RMSE changes in mmHg: estimate above descriptive 95% patient-bootstrap interval. For each patient, differences are averaged over the three matched seeds before resampling the six patients 2,000 times. Negative changes favor the method. These are not seed-SD intervals or simultaneous multiplicity-adjusted intervals. Affine calibration is unavailable below three references.

Patient	Population	Bias	Affine	Head	Last block	Full	Last reference	Reference mean
p005879	9.58 (1.99)	10.57 (3.43)	8.17 (2.26)	11.16 (3.44)	9.50 (2.44)	8.36 (2.57)	6.99 (0.00)	5.87 (0.00)
p005885	14.22 (1.69)	9.94 (0.31)	10.49 (1.44)	11.19 (1.84)	11.10 (0.89)	12.42 (2.38)	11.13 (0.00)	11.15 (0.00)
p005977	21.53 (1.52)	5.35 (0.42)	7.50 (0.81)	10.50 (1.72)	7.15 (0.23)	7.89 (0.75)	7.55 (0.00)	6.82 (0.00)
p005984	7.79 (0.03)	8.91 (0.32)	9.21 (0.45)	9.81 (0.57)	8.41 (0.70)	7.84 (0.41)	26.51 (0.00)	8.73 (0.00)
p006009	10.30 (1.91)	8.97 (0.94)	9.13 (1.00)	8.71 (0.45)	8.88 (0.81)	8.91 (0.72)	25.48 (0.00)	11.15 (0.00)
p006052	12.96 (1.49)	10.39 (1.27)	11.52 (2.49)	11.38 (1.51)	10.58 (2.71)	10.04 (4.38)	27.56 (0.00)	10.35 (0.00)

Table 34: Absolute SBP MAE in mmHg for every final patient at K=10. Cells show the mean across the three population seeds, with sample seed SD in parentheses; each cell describes one patient, not a patient confidence interval. The same later queries are used for all methods. Absolute errors expose residual failures that an improvement-only heatmap can conceal.

Patient	Population	Bias	Affine	Head	Last block	Full	Last reference	Reference mean
p005879	9.40 (2.00)	4.10 (2.05)	2.60 (0.23)	4.07 (1.79)	3.98 (1.99)	3.52 (0.64)	4.37 (0.00)	3.26 (0.00)
p005885	6.26 (0.68)	4.83 (0.64)	4.67 (0.75)	4.94 (0.83)	4.57 (0.52)	4.53 (0.84)	3.10 (0.00)	3.24 (0.00)
p005977	16.54 (2.68)	3.42 (0.10)	3.68 (0.50)	6.03 (1.56)	4.62 (0.33)	5.07 (0.95)	5.29 (0.00)	3.72 (0.00)
p005984	3.63 (0.65)	6.19 (0.28)	6.50 (0.38)	6.45 (0.80)	6.09 (0.30)	5.75 (0.35)	14.36 (0.00)	5.25 (0.00)
p006009	5.39 (1.28)	6.04 (1.21)	5.16 (0.36)	4.36 (1.21)	4.82 (1.78)	5.38 (1.80)	10.74 (0.00)	4.75 (0.00)
p006052	5.57 (2.11)	4.46 (0.62)	4.54 (1.09)	3.87 (0.59)	4.39 (0.59)	3.68 (0.24)	14.71 (0.00)	4.01 (0.00)

Table 35: Absolute DBP MAE in mmHg for every final patient at K=10. Cells show the mean across the three population seeds, with sample seed SD in parentheses; each cell describes one patient, not a patient confidence interval. The same later queries are used for all methods. Absolute errors expose residual failures that an improvement-only heatmap can conceal.

Patient	Population	Bias	Affine	Head	Last block	Full	Last reference	Reference mean
	11.48	12.34	9.99	12.87	11.35	10.11	8.34	7.45
p005879	(1.91)	(3.32)	(2.08)	(3.46)	(2.58)	(2.58)	(0.00)	(0.00)
	17.24	13.19	14.24	15.34	14.20	15.07	14.30	14.33
p005885	(1.60)	(0.55)	(2.04)	(2.75)	(1.29)	(2.47)	(0.00)	(0.00)
	22.60	6.97	9.56	12.76	9.14	10.15	9.79	8.93
p005977	(1.66)	(0.58)	(1.05)	(2.02)	(0.35)	(0.86)	(0.00)	(0.00)
	9.52	10.45	10.79	11.48	10.12	9.63	28.69	11.21
p005984	(0.25)	(0.08)	(0.16)	(0.76)	(1.00)	(0.68)	(0.00)	(0.00)
	12.45	11.11	11.30	10.88	11.11	11.03	29.29	15.28
p006009	(1.91)	(1.08)	(1.11)	(0.68)	(1.08)	(0.53)	(0.00)	(0.00)
	15.18	13.24	14.26	13.38	12.98	12.37	29.83	13.70
p006052	(1.68)	(1.86)	(3.33)	(1.71)	(3.60)	(5.08)	(0.00)	(0.00)

Table 36: Absolute SBP RMSE in mmHg for every final patient at K=10. Cells show the mean across the three population seeds, with sample seed SD in parentheses; each cell describes one patient, not a patient confidence interval. The same later queries are used for all methods. Absolute errors expose residual failures that an improvement-only heatmap can conceal.

Patient	Population	Bias	Affine	Head	Last block	Full	Last reference	Reference mean
	10.29	4.98	3.32	4.96	4.83	4.26	5.17	4.02
p005879	(1.45)	(2.17)	(0.21)	(1.84)	(2.21)	(0.65)	(0.00)	(0.00)
	7.57	6.43	6.21	6.71	6.17	6.02	4.56	4.64
p005885	(0.66)	(0.93)	(1.08)	(0.99)	(0.76)	(0.87)	(0.00)	(0.00)
	17.04	4.51	4.82	7.47	5.97	6.59	6.94	5.36
p005977	(2.69)	(0.13)	(0.61)	(1.70)	(0.50)	(1.08)	(0.00)	(0.00)
	5.09	6.92	7.28	7.32	7.06	6.76	15.27	6.71
p005984	(0.67)	(0.38)	(0.48)	(0.75)	(0.42)	(0.39)	(0.00)	(0.00)
	6.51	7.19	6.34	5.52	6.00	6.58	12.01	6.48
p006009	(1.15)	(1.36)	(0.48)	(1.43)	(2.12)	(2.13)	(0.00)	(0.00)
	6.60	5.52	5.61	4.72	5.36	4.60	15.48	4.85
p006052	(2.28)	(0.79)	(1.30)	(0.63)	(0.61)	(0.22)	(0.00)	(0.00)

Table 37: Absolute DBP RMSE in mmHg for every final patient at K=10. Cells show the mean across the three population seeds, with sample seed SD in parentheses; each cell describes one patient, not a patient confidence interval. The same later queries are used for all methods. Absolute errors expose residual failures that an improvement-only heatmap can conceal.

Strategy	K=1	K=2	K=3	K=5	K=10
Head	10^{-3} 100 steps	10^{-3} 100 steps	10^{-3} 100 steps	10^{-3} 100 steps	10^{-3} 100 steps
Last block	10^{-4} 100 steps	10^{-4} 100 steps	10^{-4} 100 steps	10^{-4} 100 steps	10^{-4} 100 steps
Full	10^{-4} 20 steps	10^{-4} 20 steps	10^{-4} 20 steps	10^{-4} 20 steps	10^{-4} 100 steps
Affine	–	–	$\lambda = 10$	$\lambda = 1$	$\lambda = 1$

Table 38: Actual frozen development-selected settings. Neural cells show learning rate above update steps; affine cells show the slope-shrinkage penalty. One setting per strategy and budget was chosen using four separate development patients averaged across three seeds, and then used unchanged for every final target. No final-target query determined these settings.

Model (runs)	CPU, B=1	CPU, B=256	MPS, B=1	MPS, B=256
CNN (3)	0.896 [0.784, 0.913]	232.038 [164.191, 321.185]	2.014 [1.813, 2.287]	4.048 [3.564, 5.716]
SE-last (3)	1.298 [0.477, 1.481]	282.591 [173.820, 306.271]	2.365 [1.851, 3.438]	3.446 [3.385, 3.881]
Temporal (3)	1.263 [0.851, 1.423]	349.588 [179.594, 377.307]	3.349 [1.836, 4.739]	4.392 [3.287, 9.987]
CBAM1D (3)	1.124 [0.948, 1.137]	262.773 [225.993, 284.531]	3.218 [2.905, 4.725]	5.089 [4.227, 7.355]
Additive (3)	0.640 [0.399, 1.077]	283.914 [147.819, 288.189]	3.473 [1.474, 5.670]	7.745 [6.145, 13.561]
SE-all (3)	0.683 [0.558, 0.749]	242.903 [216.187, 353.065]	4.216 [1.628, 4.666]	10.042 [7.842, 22.685]
ECG only (1)	0.866	313.172	3.916	8.003
PPG only (1)	0.520	301.540	4.319	8.162

Table 39: Measured warm forward-pass batch latency in milliseconds. Each run times 20 repetitions after five warmups with device synchronization. Cells show the median of run-specific medians, with the range across runs below; single-seed ablations show their measured median. B=256 entries are whole-batch times, not per-window latency (divide by 256 for amortized time per example). Inputs already reside on the measured device; these exclude signal acquisition, preprocessing and input/output transfers, and are not Raspberry Pi timings. Run-to-run timing spread is measurement variability, not an accuracy confidence interval; unavailable devices are marked –.

Patient	Pressure	Query N	Support mean (SD)	Query mean (SD)	Query − support
p005879	SBP	654	92.9 (6.0)	94.6 (7.3)	+1.6
p005879	DBP	654	48.2 (3.0)	46.6 (3.7)	-1.5
p005885	SBP	1220	135.4 (22.8)	130.4 (13.4)	-5.0
p005885	DBP	1220	73.1 (6.9)	72.2 (4.6)	-0.9
p005977	SBP	569	109.5 (20.1)	106.5 (8.4)	-3.0
p005977	DBP	569	50.3 (8.1)	51.8 (5.1)	+1.5
p005984	SBP	666	97.2 (12.7)	98.1 (11.2)	+0.9
p005984	DBP	666	46.3 (7.9)	50.4 (5.3)	+4.1
p006009	SBP	761	104.7 (13.0)	109.7 (14.5)	+5.0
p006009	DBP	761	51.2 (5.7)	54.8 (5.4)	+3.6
p006052	SBP	414	91.0 (23.3)	97.4 (12.1)	+6.4
p006052	DBP	414	54.3 (9.4)	54.4 (4.8)	+0.1

Table 40: Reference-pressure context for all six final patients at K=10, using the saved seed-42 population artifact once per patient. Pressure values and mean shifts are in mmHg; each support set has ten references. SD uses $n - 1$. Signed shifts are later query mean minus early support mean. Full minima/maxima and elapsed spans are retained in the companion JSON. These are reference distributions, not prediction scores.

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